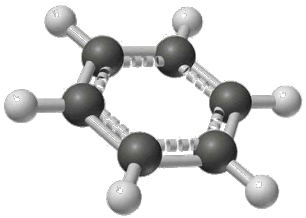
CHEM 1043 General Chemistry II

Topic 2 Structure and Bonding – The Delocalized Approach: Molecular Orbital Theory

Prof. Seungkyu Lee



What to learn in this topic?



Structure and Bonding – The Delocalized Approach:

1. Molecular orbital (MO) theory for diatomic molecules
2. Delocalized electrons: MO theory for larger molecules
3. Band theory for metals, semiconductors, and insulators



Your reading:

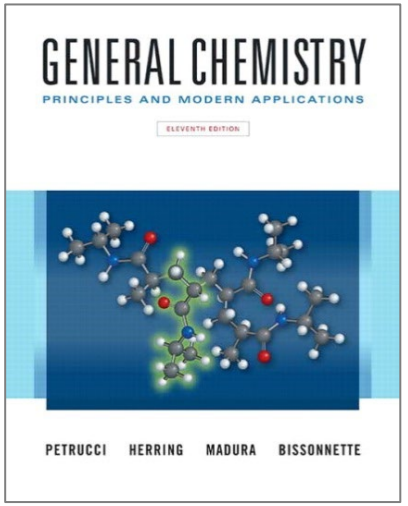
11-5 – 11-7, Chapter 11)**

Chemical Bonding II:

Valence Bond and Molecular Orbital Theories

Petrucci; Herring; Madura; Bissonnette:

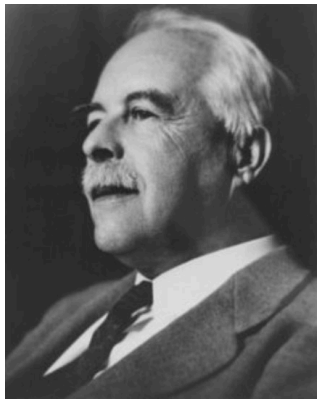
General Chemistry: Principles and Modern Applications, 11th edition, Pearson



** *Note that 11-1 – 11-4 have been covered in CHEM1042.*

Recall what you have learnt in CHEM1042 (1):

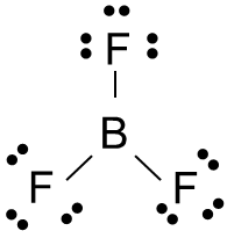
Bonding Theories:

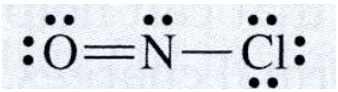


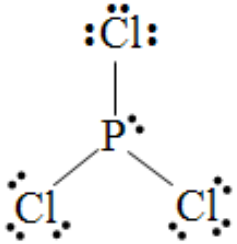
1. Lewis theory (Gilbert N. Lewis, 1875-1946)

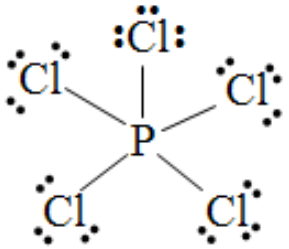
   (Lewis structure tells the atomic connectivity in a substance.)

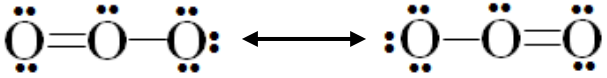
For example:











Recall what you have learnt in CHEM1042 (2):

2. VSEPR theory

(It is good at predicting molecular shapes.)

Electron pairs repel each other, whether they are in chemical bonds (bond pairs) or unshared (lone pairs). Electron pairs assume orientations about an atom to minimize repulsions.

No. of electron group	**Electron group geometry**	No. of lone pairs	**Molecular geometry**	Example (bond angle)
2	Linear	0	Linear	$BeCl_2$ (180°)
3	Trigonal planar	0	Trigonal planar	BF_3 (120°)
		1	Bent	SO_2 (119°)
4	Tetrahedral	0	Tetrahedral	CH_4 (109.5°)
		1	Trigonal pyramidal	NH_3 (107°)
		2	Bent	H_2O (104.5°)
5	Trigonal bipyramidal	0	Trigonal bipyramidal	PCl_5 (90°, 120°)
		1	Seesaw	SF_4 (89.5°, 103°)
		2	T-shaped	ClF_3 (87.5°)
		3	Linear	XeF_2 (180°)
6	Octahedral	0	Octahedral	SF_6 (90°)
		1	Square pyramidal	BrF_5 (84.8°)
		2	Square planar	XeF_4 (90°)

Recall what you have learnt in CHEM1042 (3):

3. Valence bond theory

(It predicts how a covalent bond is formed in terms of atomic orbital (AO) overlap.)

TABLE 11.1 Some Hybrid Orbitals and Their Geometric Orientations

Hybrid Orbitals	Geometric Orientation	Electron group geometry	Example
sp	Linear		$BeCl_2$
sp^2	Trigonal-planar		BF_3
sp^3	Tetrahedral		CH_4
sp^3d	Trigonal-bipyramidal		PCl_5
sp^3d^2	Octahedral		SF_6

Lewis theory, VSEPR theory and VB theory provide satisfactory explanations and predictions to molecular structure in most cases, but still fail to explain some aspects in bonding.

E.g.) ***Why does O_2 interact with a magnetic field?***

Why are some molecules coloured?

Information about **Magnetic Properties**:

- Two common types of magnetic behaviour:

(a) Diamagnetic – no unpaired electron

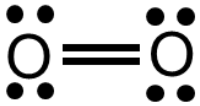
When two spins are opposing, the magnetic fields cancel each other. Diamagnetic substances are weakly repelled by external magnetic fields.

(b) Paramagnetic – with unpaired electron(s)

When spins are unpaired, the magnetic fields do not cancel. Paramagnetic substances are attracted by external magnetic fields.



Consider O_2,



- Diatomic oxygen O_2 has an even number of valence electrons (12).
- By Octet rule, all valence electrons are paired → O_2 should be diamagnetic.



- Experiments show that liquid oxygen is attracted by the magnetic field → O_2 must have unpaired electrons (i.e. *paramagnetic*).
- Explanation is needed from an alternative bonding theory:

Molecular Orbital Theory

- So far, The best theory to explain the various properties of molecules and materials

1. Molecular Orbital Theory

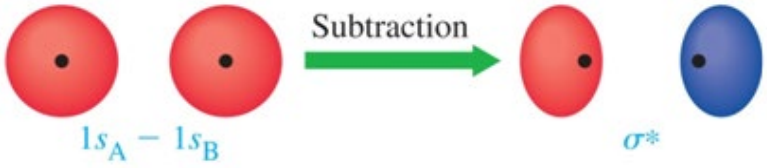
- Molecular orbital (MO) theory treats bonding in terms of **molecular orbitals** that can extend over an entire molecule.

- A molecule is seen as a collection of nuclei and electrons, where the electrons are assumed to occupy orbitals much as they do in atoms, but having the orbitals extend over the entire molecule.

- The electrons are assumed to be ***delocalized*** rather than always located between a given pair of atoms.



$$\Psi_1 (\sigma_{1s}) = \varphi_1 (1s_A) + \varphi_2 (1s_B)$$



$$\Psi_2 (\sigma^*_{1s}) = \varphi_1 (1s_A) - \varphi_2 (1s_B)$$

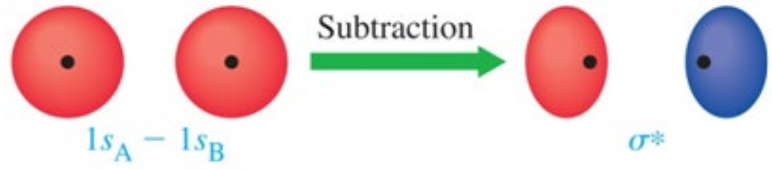
1. Molecular Orbital Theory

- Recall that orbitals are wave functions. As two atoms approach each other, the wave functions of their atomic orbitals (AOs) combine to form a set of new molecular orbitals (MOs). This is the linear combination of atomic orbitals (LCAO).

Each MO can be represented mathematically as a combination of AOs.



$$\Psi_1 (\sigma_{1s}) = \varphi_1 (1s_A) + \varphi_2 (1s_B)$$



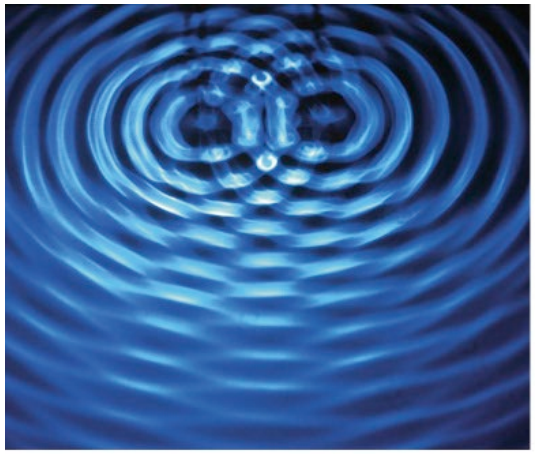
$$\Psi_2 (\sigma^*_{1s}) = \varphi_1 (1s_A) - \varphi_2 (1s_B)$$

Recall what you have learnt in "Atoms: the quantum world" in CHEM1042

Quantum Mechanics:

- Energy in atoms and molecules are quantized
- Matter and energy have a dual nature – part particle and part wave.

- Recall that orbitals are wave functions, mathematical solutions of the Schrödinger wave equation. ψ^2 is a quantity related to probability density distributions.



When two waves intersect →

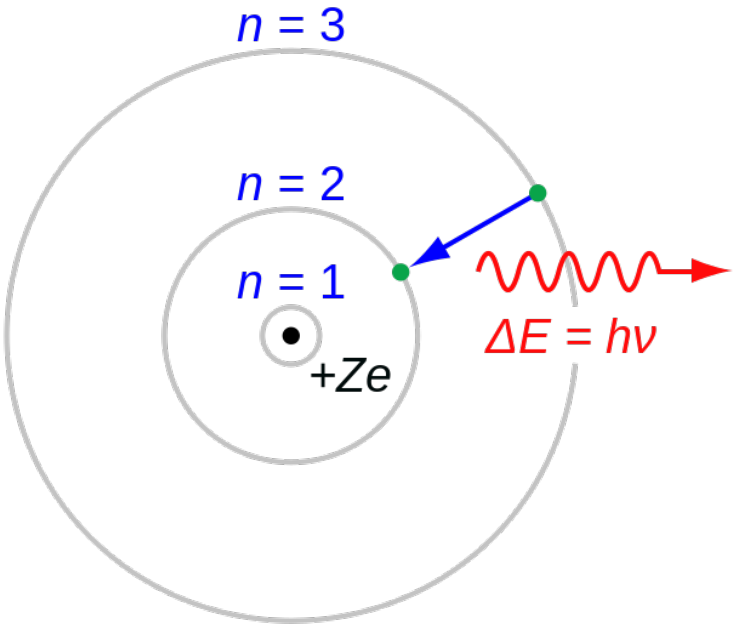
- some places where the waves disappear (***destructive interference***, waves are *out of phase*) and
- some places where the waves persist (**constructive interference**, waves are *in phase*).

Quantum Mechanics:

- Energy in atoms and molecules are quantized

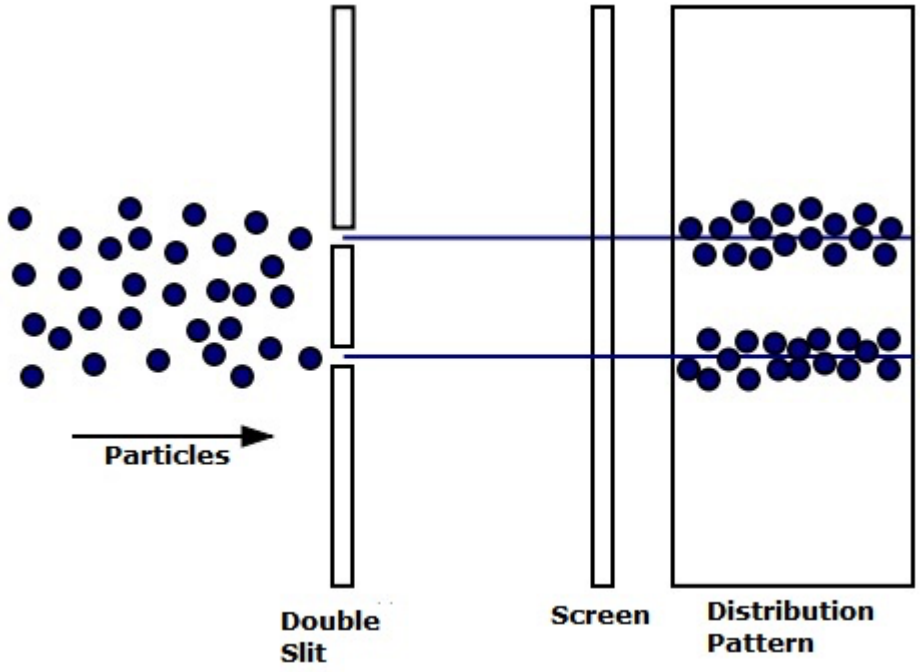
Bohr model (1913)

Electrons are particles and orbit around the nucleus



Double slit experiment (1927)

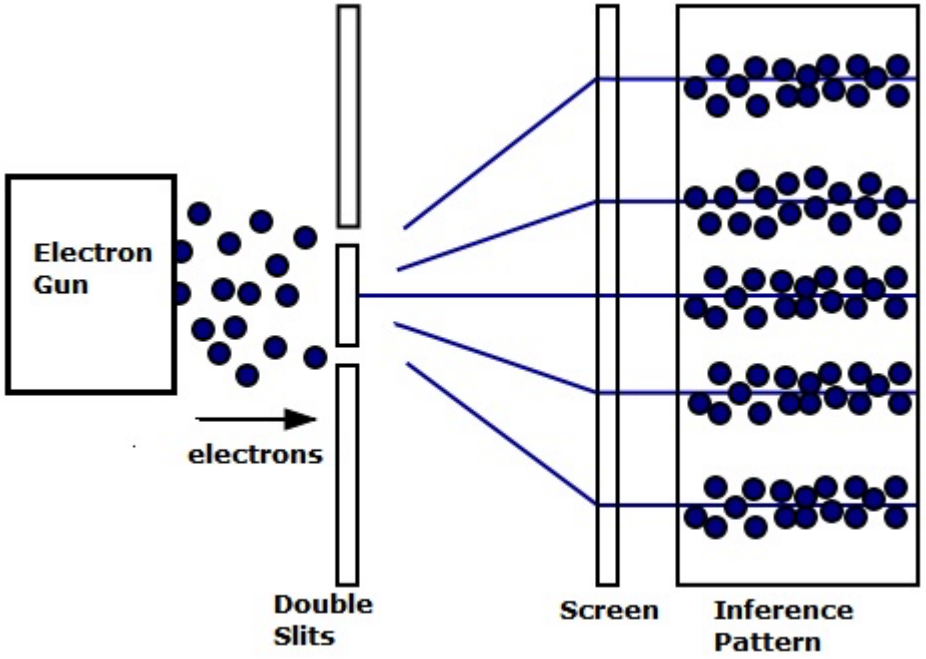
If electrons are particles,

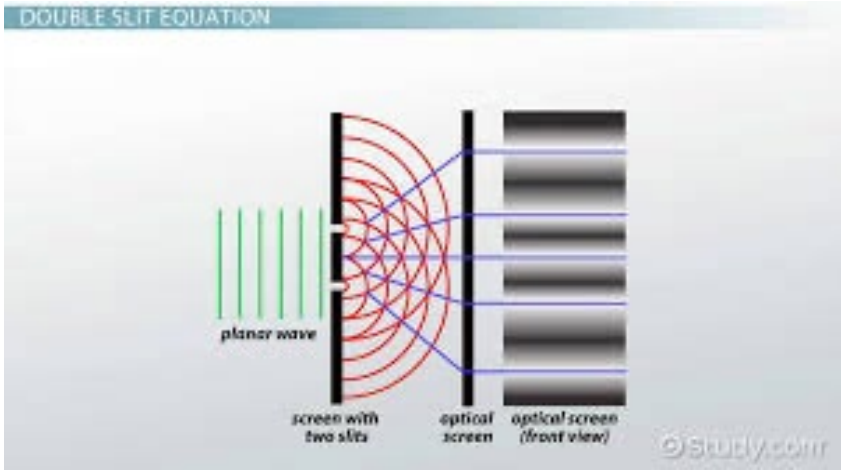


Quantum Mechanics:

Matter and energy have a dual nature – part particle and part wave.

However, the result is

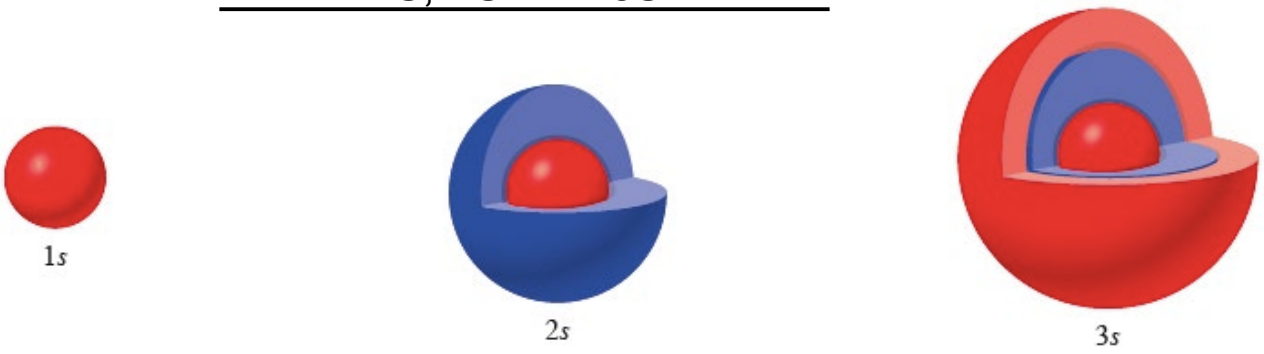




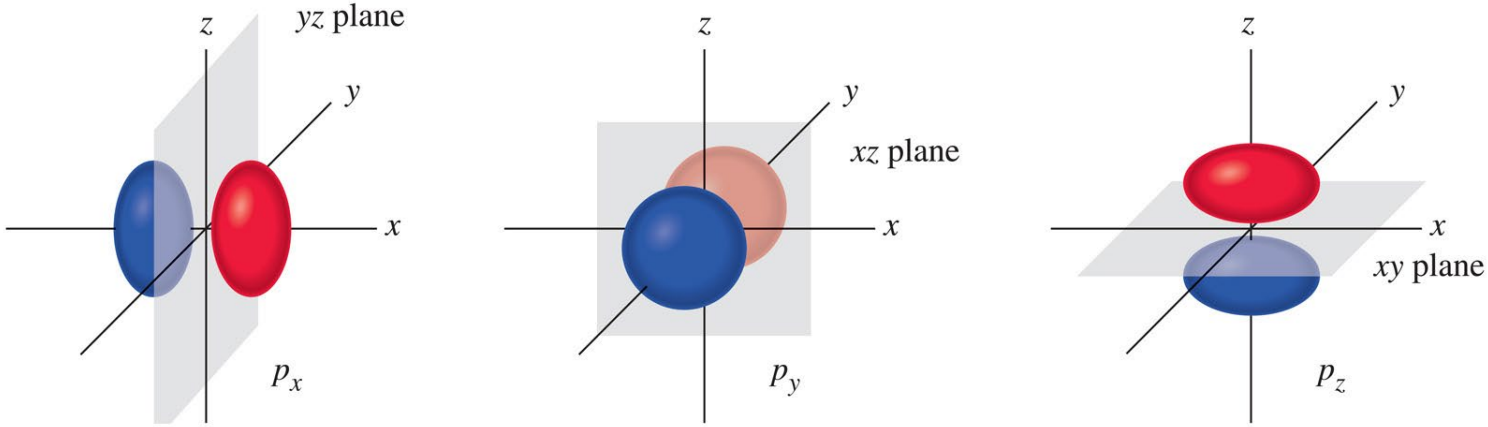
Electrons have wave-like property also!

Recall what you have learnt in "Atoms: the quantum world" in CHEM1042

Three-dimensional representations of the 95% electron probability density for the **1*s***, **2*s*** and **3*s*** orbitals

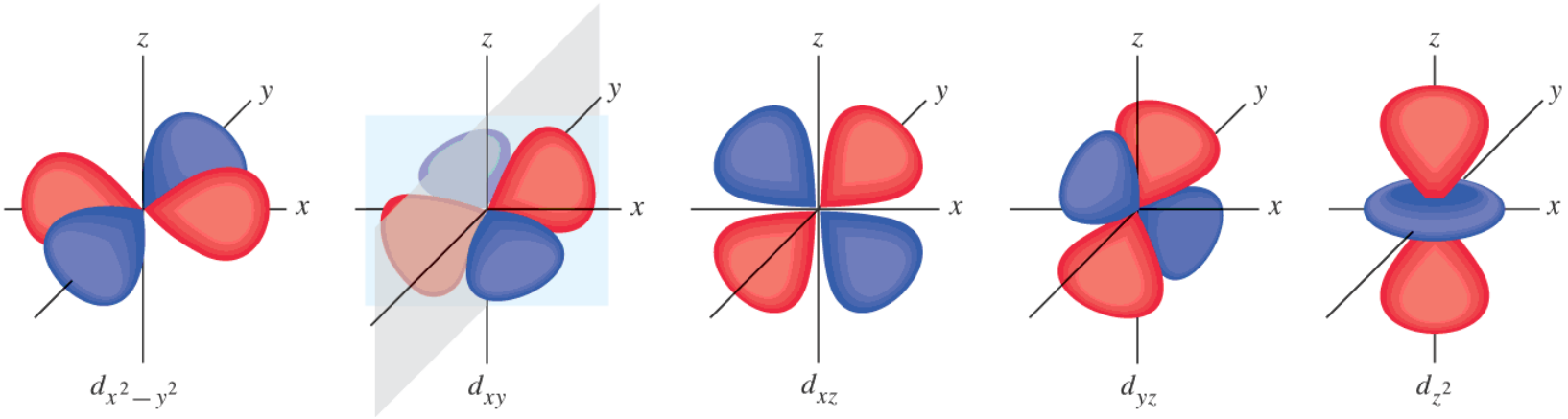


Three-dimensional representations of the 95% electron probability density for the three **2*p*** orbitals



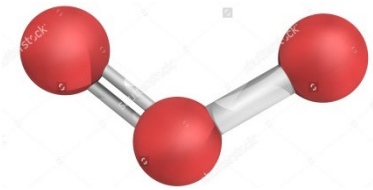
Recall what you have learnt in "Atoms: the quantum world" in CHEM1042

Three-dimensional representations of the 95% electron probability density for the five ***d*** orbitals

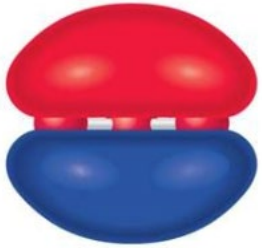


The Molecular Orbital Approach

The wave functions of electrons initially localized on different atoms begin to interact with one another and form new wave functions that represent chemical bonds in molecules (i.e. the new wave functions are ***delocalized***).

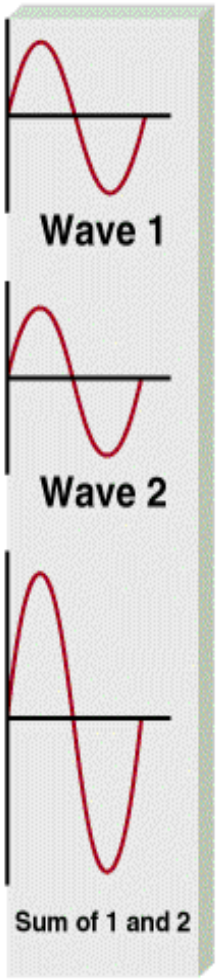


The ball-and-stick model of the ozone molecule.
Localized model based on Lewis and Valence bond theories

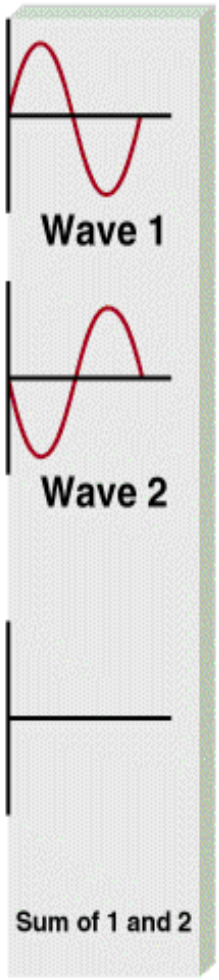


The electrons are actually not localized in pairs to form covalent bonds between the nuclei.
Delocalized model based on MO Theory

Constructive



Destructive



Molecular Orbital Theory

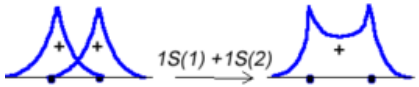
- Consider the formation of H_2, when the two H atoms approach, the two 1*s* wave functions combine by interfering constructively or destructively.

(a) Constructive interference

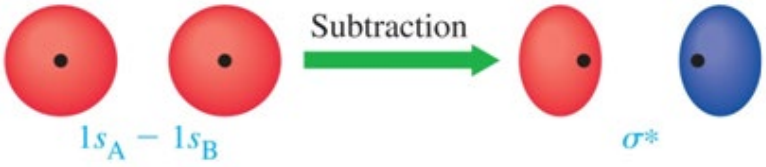


$\Psi_1 (\sigma_{1s}) = \varphi_1 (1s_A) + \varphi_2 (1s_B)$

The positive sign puts the waves in phase.

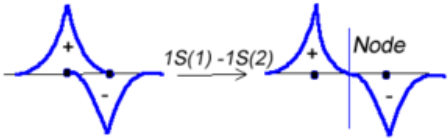


(b) Destructive interference



$\Psi_2 (\sigma^*_{1s}) = \varphi_1 (1s_A) - \varphi_2 (1s_B)$

The negative sign puts the waves out of phase.



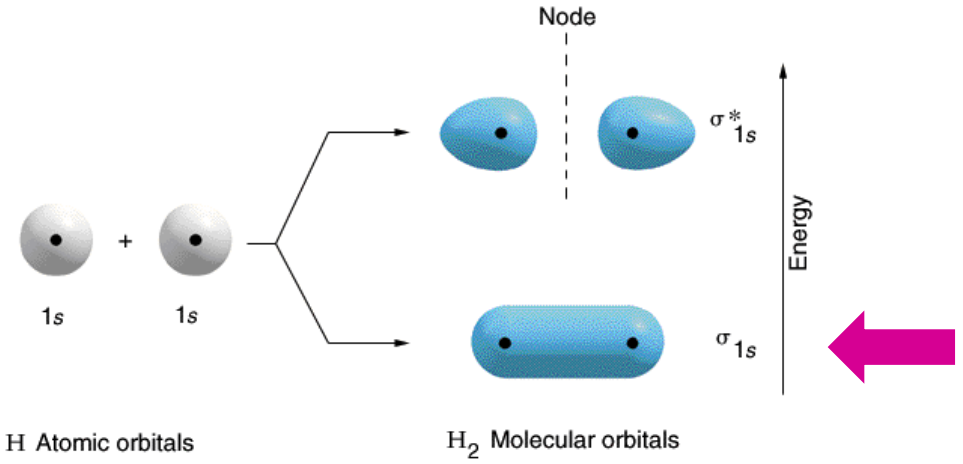
(a) The addition of two 1*s* orbitals in phase to form a **σ_{1s} molecular orbital**. This orbital produces electron density between the nuclei, leading to a chemical bond.

(b) The addition of two 1*s* orbitals out of phase to produce a **σ^*_{1s} antibonding orbital**. This orbital has a nodal plane perpendicular to the inter-nuclear axis, as do all antibonding orbitals.

(a) Constructive interference



$$\Psi_1 (\sigma_{1s}) = \varphi_1 (1s_A) + \varphi_2 (1s_B)$$

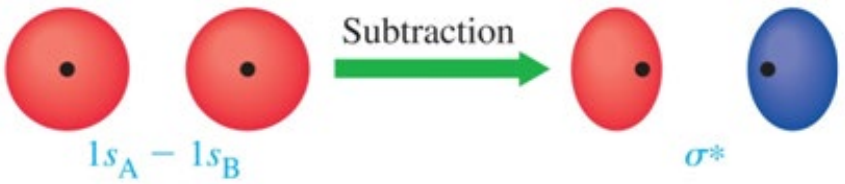


- The **constructive interference** of the two wave functions leads to a ***greater*** **electron probability** between the nuclei. The increased electron charge density between the nuclei causes them to draw closer together, forming a chemical bond.

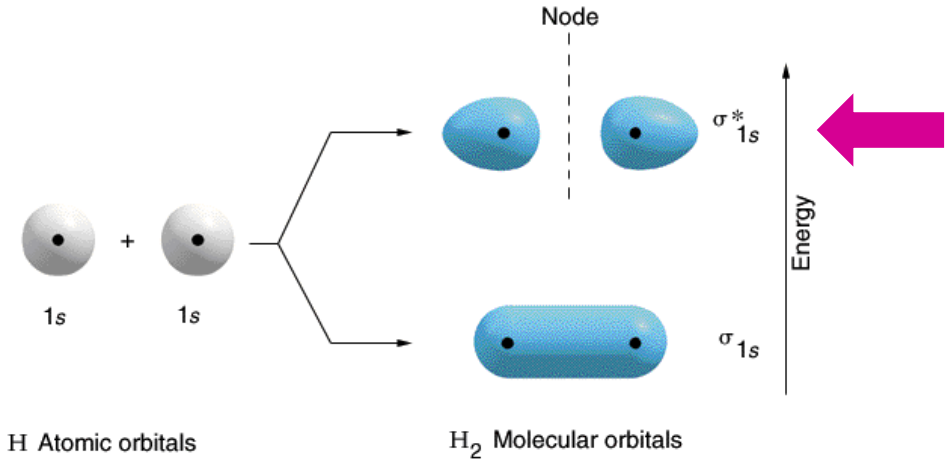
- The electron probability in σ_{1s} **orbital** is $(1s_A + 1s_B)^2$, where $1s_A + 1s_B$ are the two 1*s* orbitals on the two H atoms. $2 \times 1s_A 1s_B$ is the extra charge density between the nuclei → the **bonding molecular orbital** is the result because there is a **high electron charge density** between the two nuclei.

Recall that ψ^2 is the electron probability density.

(b) Destructive interference

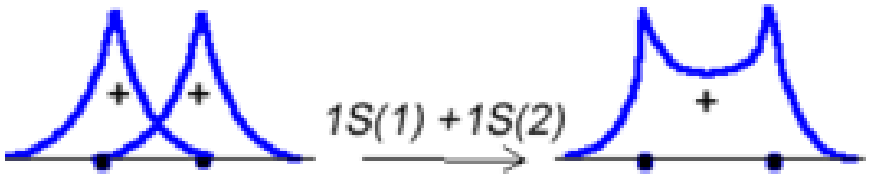


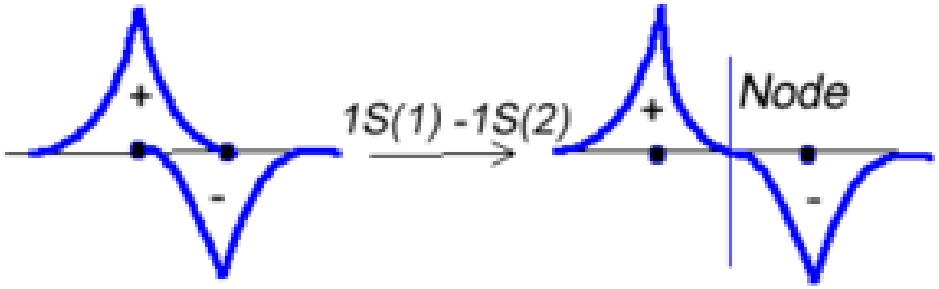
$$\Psi_2 (\sigma^*_{1s}) = \varphi_1 (1s_A) - \varphi_2 (1s_B)$$

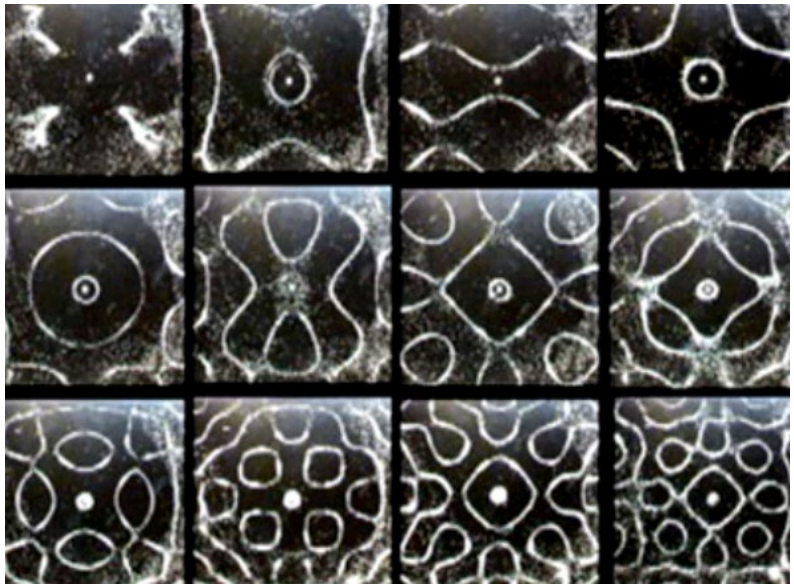


- The molecular orbital formed by the **destructive interference** of the two 1*s* orbitals leads to ***reduced*** **electron probability** between the nuclei. This produces an **antibonding molecular orbital** (designated by *) because it places a very low electron charge density between the two nuclei. $-2 \times 1s_A 1s_B$ is the loss of charge density between the nuclei.

- With a low electron charge density between the two nuclei in σ^*_{1s} orbital, strong repulsion occurs and the bond is weakened, hence the term “antibonding”.


+
+
1S(1) +1S(2)
+

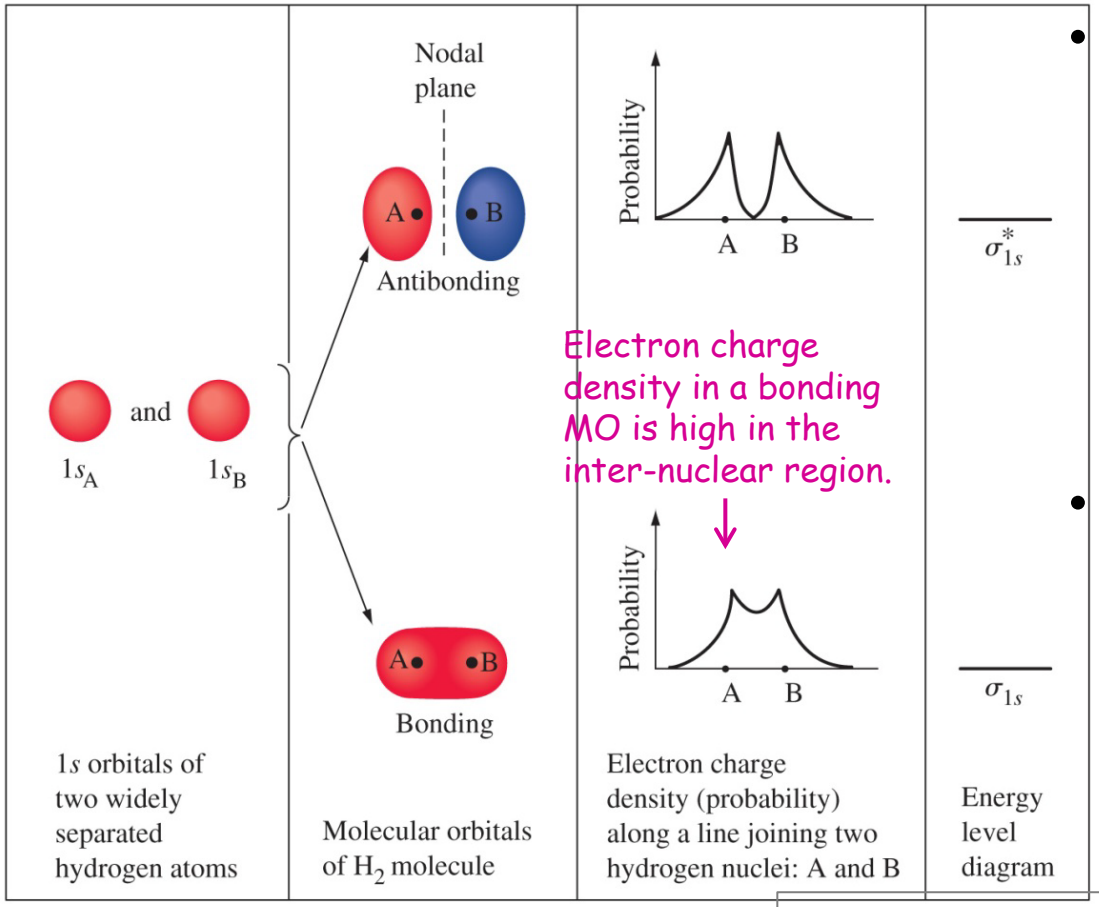

+
-
1S(1) -1S(2)
+
Node
-





Molecular Orbital Theory

Fig. 11-22) The interaction of two hydrogen atoms according to molecular orbital theory



a plane of zero amplitude

- σ^*_{1s} MO (H_2): higher energy level than 1*s* AOs (H) → Filling e^- into antibonding MO σ^*_{1s} brings **destabilization** for H_2 formation.

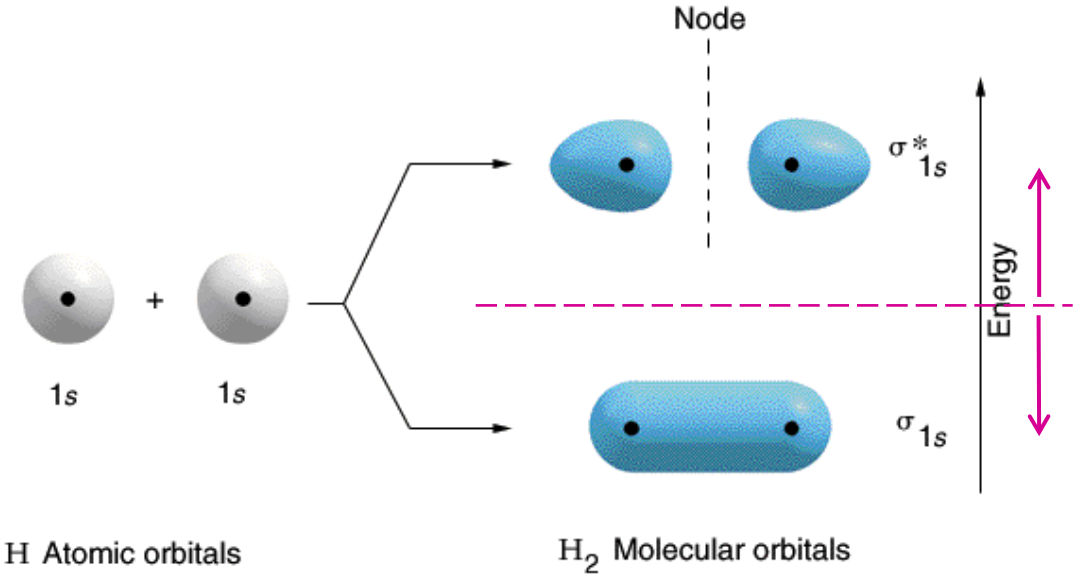
- σ_{1s} MO (H_2): lower energy level than 1*s* AOs (H) → Filling e^- into bonding MO σ_{1s} brings **stabilization** for H_2 formation.

- All antibonding orbitals have a nodal plane perpendicular to the inter-nuclear axis.

Quick Summary on H_2 molecule

- In H_2, the MOs resulted from head-on overlapping of two 1*s* orbitals are σ_{1s} (bonding) and σ^*_{1s} (antibonding) MOs.

- Sigma (σ / σ^*) MOs are characterized by the centered electron density along the axis passing through the two nuclei.

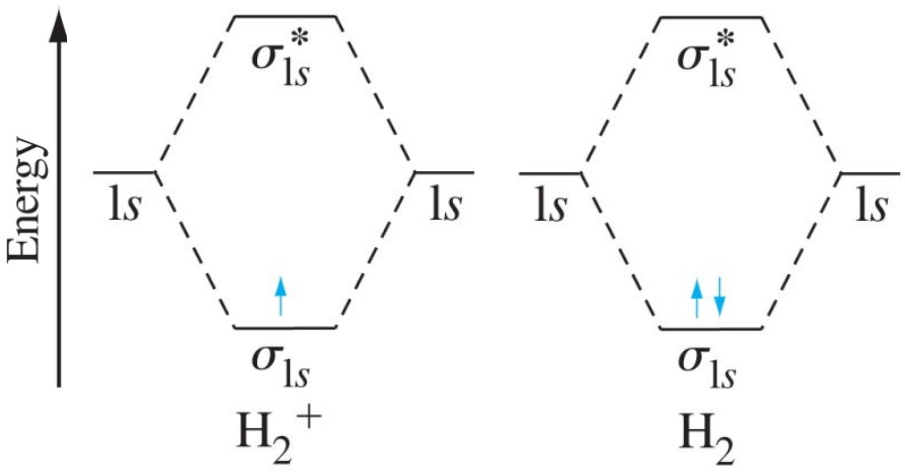


Basic Ideas Concerning MOs

1. Number of AOs combined = Number of MOs formed
2. Combination of two AOs forms two MOs, one bonding (lower energy) and one antibonding (higher energy).
3. Only AOs of similar energy and symmetry combine to form MOs.
4. Electrons fill MOs from low to high energies (Aufbau Process).
5. Each MO can accommodate up to two electrons (follow Pauli Exclusion Principle).
6. Electrons fill degenerate orbitals singly before pairing up (follow Hund's Rule).
7. $$\text{Bond Order} = \frac{\text{no. of } e^- \text{ in bonding MOs} - \text{no. } e^- \text{ in antibonding MOs}}{2}$$

Diatomic Molecules of the 1st Period Elements

- **For Hydrogen (H: $1s^1$):**



- H_2^+: one e⁻ in σ_{1s}

$$BO_{H_2^+} = \frac{(1-0)}{2} = \frac{1}{2}$$

a half bond

- H_2: two e⁻ in σ_{1s}

$$BO_{H_2} = \frac{(2-0)}{2} = 1$$

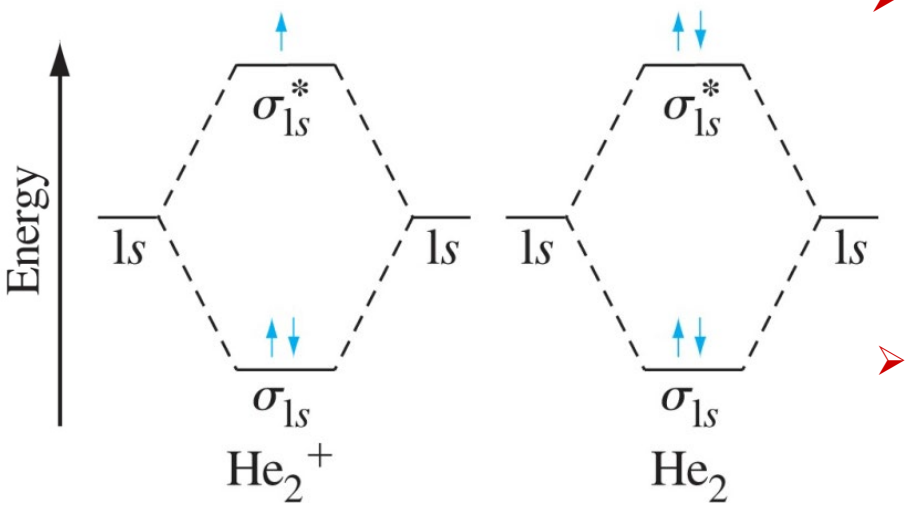
a single bond

Molecular orbital (MO) diagrams for the diatomic molecules and ions of the first-period elements

The energy levels of the isolated atoms are shown to the left and right of each diagram. The line segments in the middle represent the molecular orbital energy levels. Note that σ_{1s} is lower than 1*s* levels and σ^*_{1s} is higher than the 1*s* levels.

Diatomic Molecules of the 1st Period Elements

- **For Helium (He: $1s^2$):**



- **He_2^+:** two e^- in σ_{1s} & one e^- in σ^*_{1s}

$$BO_{He_2^+} = \frac{(2-1)}{2} = \frac{1}{2}$$

a half bond

- **He_2:** two e^- in σ_{1s} & two e^- in σ^*_{1s}

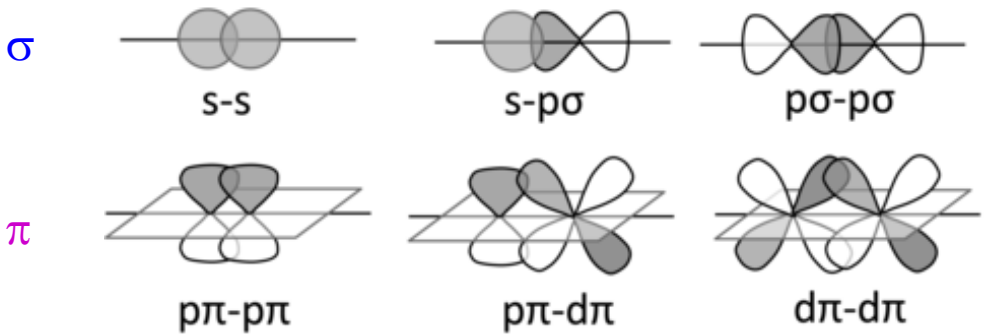
$$BO_{He_2} = \frac{(2-2)}{2} = 0$$

no bond (∴not a stable species)

Diatomic Molecules of the 2nd Period Elements

- MO formation in the 2nd period elements involves 2*s* and 2*p* orbitals.
- The MOs formed by combining 2*s* AOs are similar to those from 1*s* AOs, except they are at a higher energy.
- ***Two*** possible ways for ***2p*** orbitals combination (σ and π bonds),.

2-fold rotation along the bond axis
→ Signs don't change: σ
→ Signs change: π

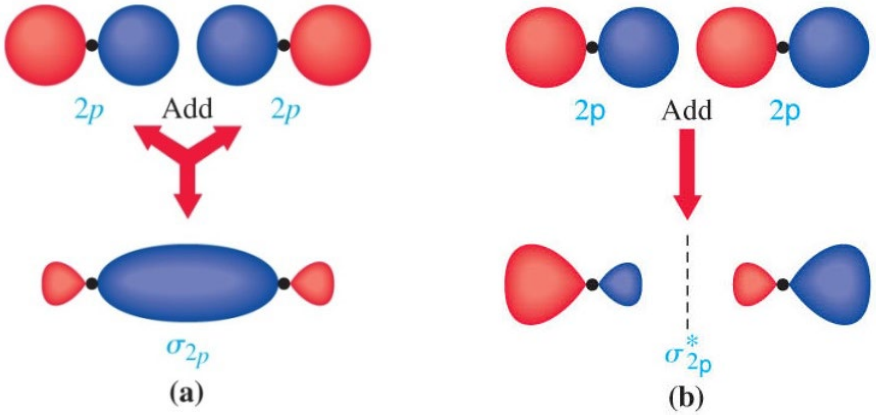


Diatomic Molecules of the 2nd Period Elements

Fig. 11-24

Formation of bonding and antibonding MOs from 2*p* orbitals

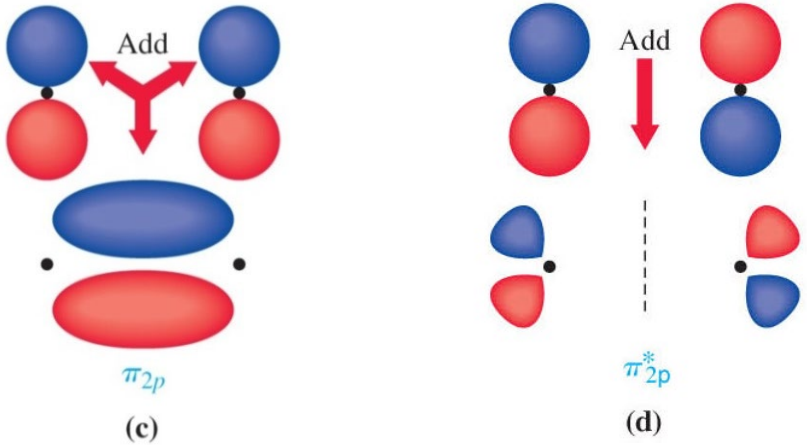
Addition along the inter-nuclear axis



(a) head-on overlap (orbitals in phase)
→ ↑ electron density in the inter-nuclear region and a σ_{2p} orbital is produced.

(b) head-on overlap (orbitals out-of-phase)
→ **a nodal plane** midway between the nuclei is formed and a σ^*_{2p} orbital is produced.

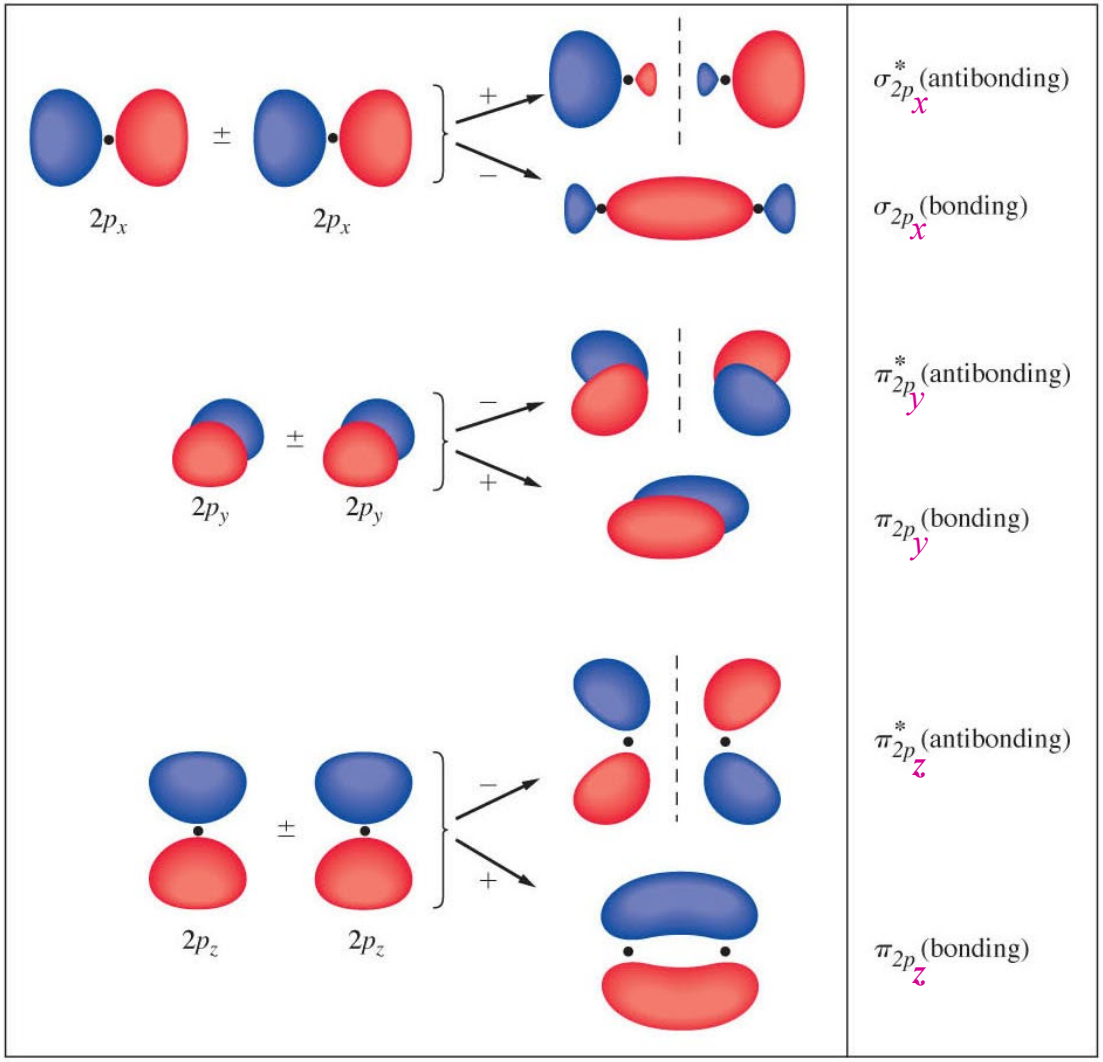
Addition perpendicular to the inter-nuclear axis



(c) sideway overlap (orbitals in phase)
→ additional electron density between the nuclei and a π_{2p} orbital is produced.

(d) sideway overlap (orbitals out-of-phase)
→ **a nodal plane** between the nuclei is formed and a π^*_{2p} orbital is produced.

The electron charge distributions of MOs formed by the 2*p* orbitals interaction



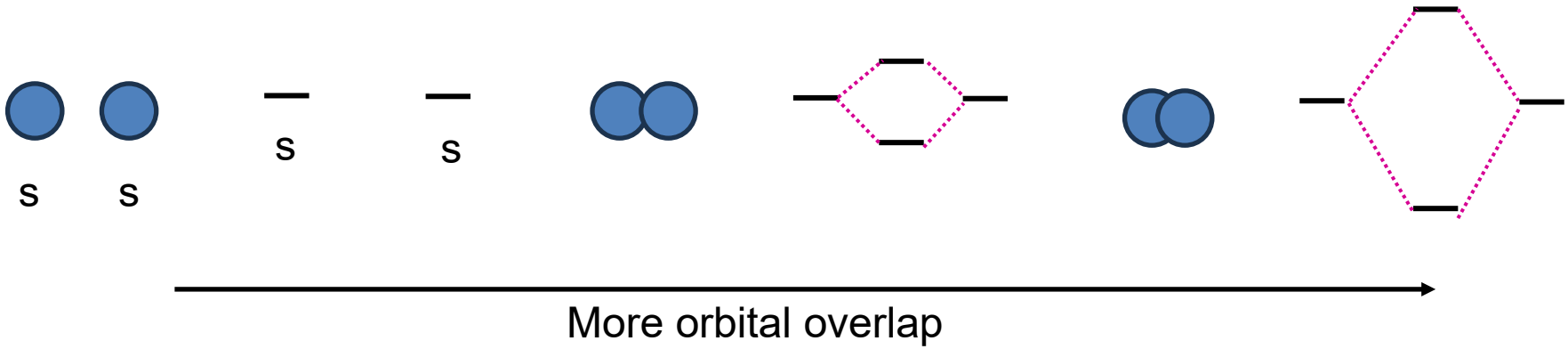
Electron charge distributions

- **Bonding** MOs place a high electron density between the nuclei.
- In **antibonding** MOs, there are **nodal planes** (denoted by dashed lines) between the nuclei, where the electron density falls to zero.

Summary of the MO diagram for the diatomic molecules of 2nd period elements:

- 1*s* AO combines with another 1*s* AO → σ_{1s} / σ^*_{1s} MOs
- 2*s* AO combines with another 2*s* AO → σ_{2s} / σ^*_{2s} MOs
- $2p_x$ AO combines with another $2p_x$ AO → σ_{2p} / σ^*_{2p} MOs
- $2p_y$ ($2p_z$) AO combines with another $2p_y$ ($2p_z$) AO → Two pairs of doubly degenerate π_{2p} / π^*_{2p} MOs

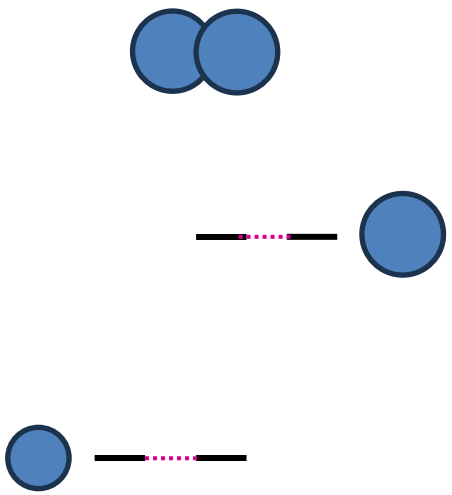
- Note that σ-type bonding orbitals should be of lower energy than π-type bonding orbitals because head-on overlap is more extensive than sideway overlap.



Summary of the MO diagram for the diatomic molecules of 2nd period elements:

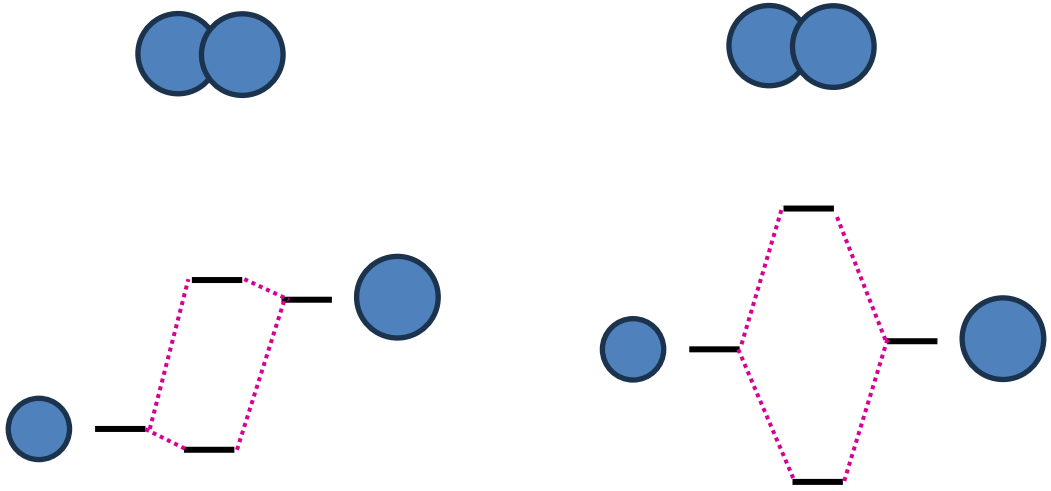
- 1s and 2s orbitals do not combine.
- σ and π orbitals do not combine.

Only AOs of similar energy and symmetry combine to form MOs.



Non bonding orbitals

Not similar energy

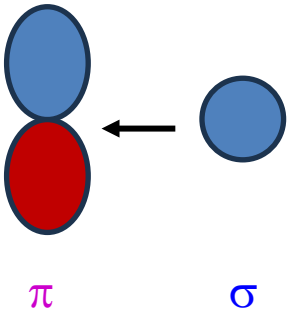


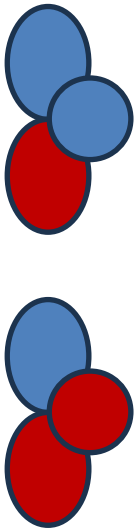
Bonding and anti bonding orbitals

Summary of the MO diagram for the diatomic molecules of 2nd period elements:

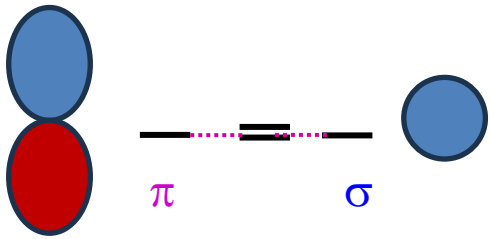
- 1s and 2s orbitals do not combine.
- σ and π orbitals do not combine.

Only AOs of similar energy and symmetry combine to form MOs.





Not similar symmetry



Non bonding orbitals

Summary of the MO diagram for the diatomic molecules of 2nd period elements:

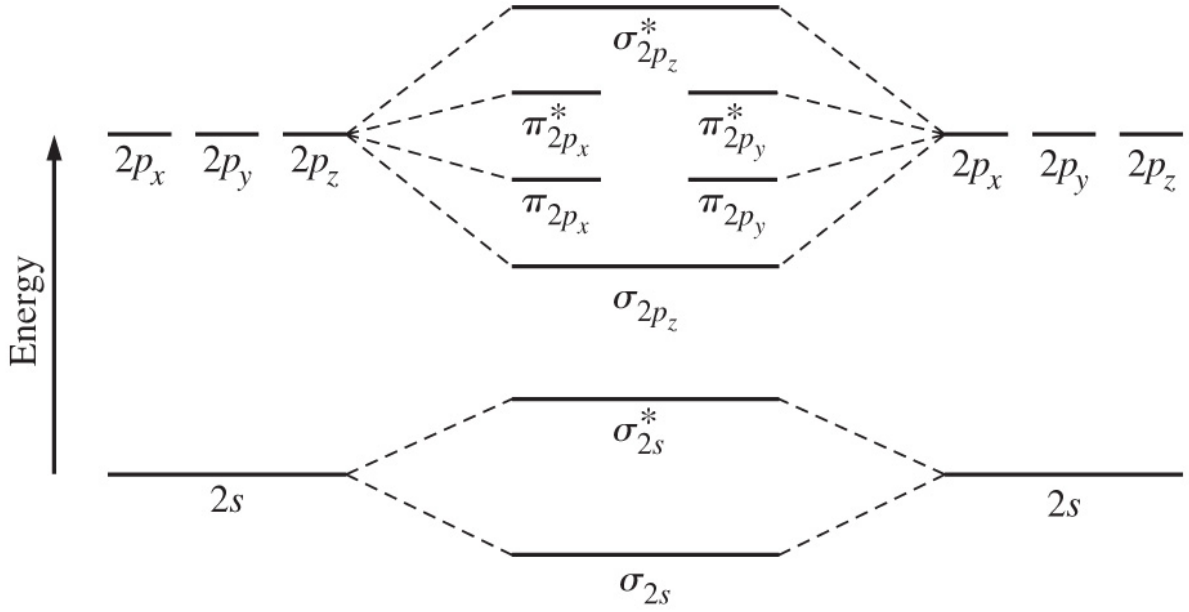
- All 2nd period elements have atomic electron configuration of $1s^2$. Therefore, when forming MOs they all have the σ_{1s} / σ^*_{1s} filled.
- Core electrons are usually ignored in MO diagrams.

∵ net bonding is determined by the interaction between the valence atomic orbitals.

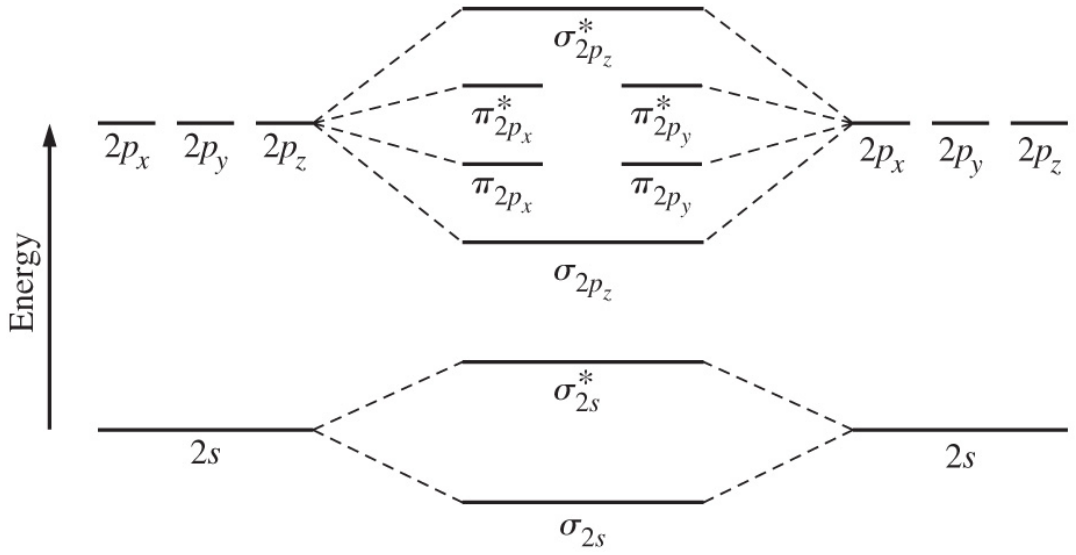
Homonuclear Diatomic Molecules of the 2nd Period Elements

- The following energy level diagram has **one assumption**:
  - *s* orbitals mix only with *s* orbitals and **2s–2s orbital interaction**
  - *p* orbitals mix only with *p* orbitals **2p–2p orbital interaction**

Fig. 11-25a)
MO energy level diagram for diatomic molecules of 2nd period elements



O, F, Ne $Z \geq 8$



- 2*s* AO lower in energy than 2*p* AO *(energy difference is large and no interaction in this case)*

  → σ_{2s} / σ^*_{2s} MOs are lower in energy than σ_{2p} / σ^*_{2p} MO.

- Overlapping efficiency: Higher in σ than in π

  → σ_{2p} is lower in energy than π_{2p};

  σ^*_{2p} is higher in energy than π^*_{2p}.

- True for **Z ≥ 8 (O, F, Ne)**, but invalid for Z ≤ 7 (Li to N).

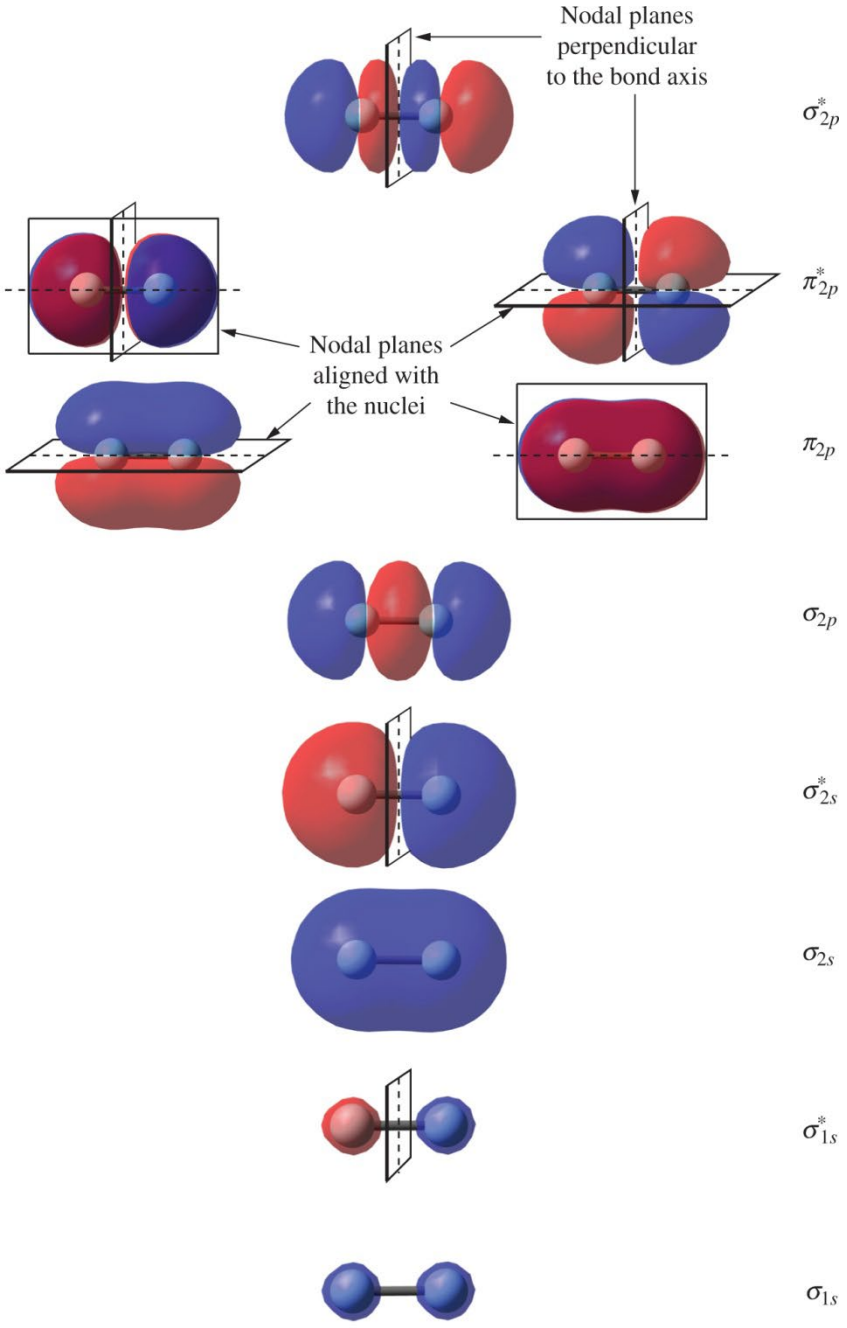
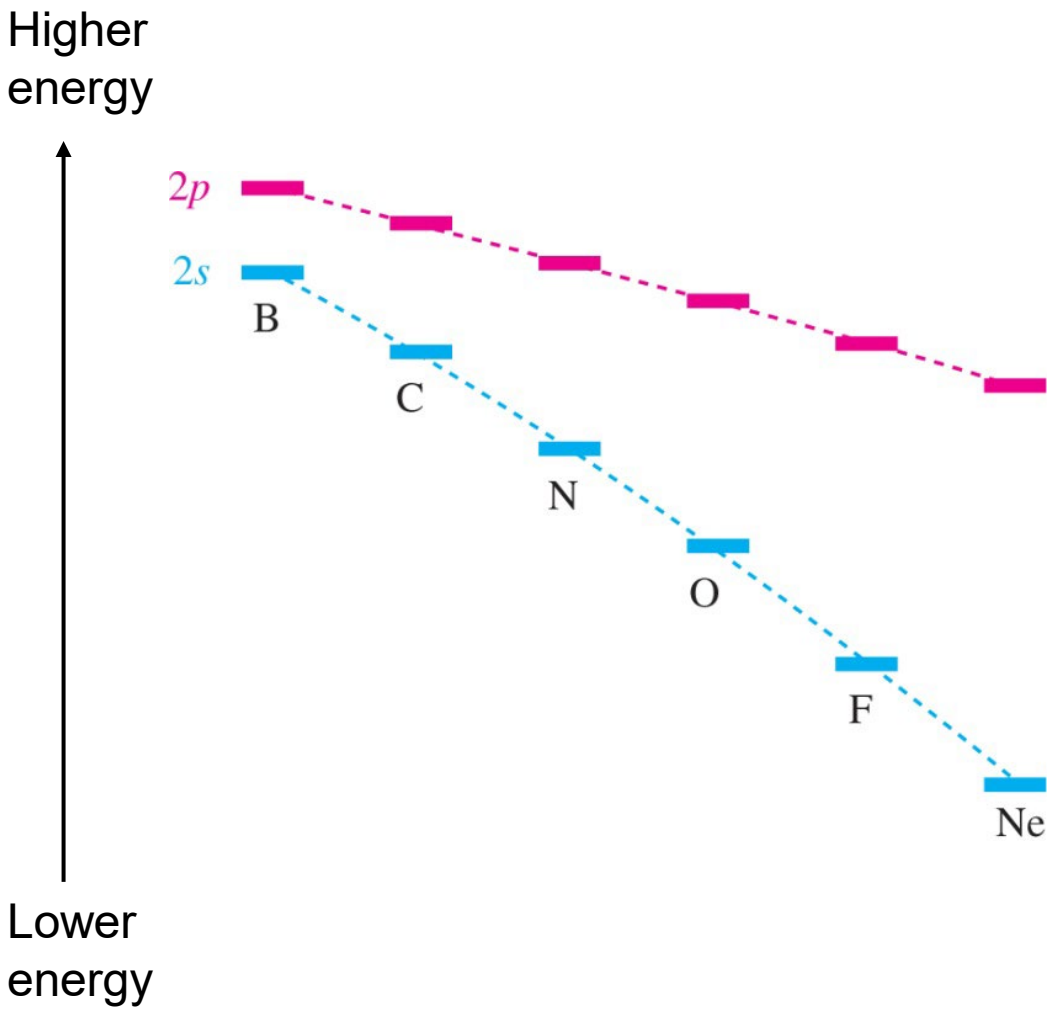


Fig. 11-21) Computed molecular orbitals of F_2

- The molecular orbitals of F_2 can be classified as **σ** or **π** and as bonding or antibonding.
- The classification depends on the presence or absence of certain types of nodal planes. The symbol π indicates the presence of a nodal plane aligned with the two nuclei whereas the symbol σ indicates the absence of such a nodal plane. An antibonding orbital is labeled with an asterisk (*) and is characterized by the presence of a nodal plane perpendicular to and between the two nuclei. In a bonding orbital, there is no such nodal plane.
- The subscript on σ or π is a reference to the atomic orbitals (one on each atom) that are combined mathematically in the LCAO method to form a particular molecular orbital.
- For example, the σ_{2p} orbital is represented mathematically as a linear combination of 2p orbitals, one on each F atom. The formation of bonding and antibonding orbitals from 1s and 2p orbitals is illustrated in Figs. 11-22 and 11-24.


Higher energy
2p
2s
B
C
N
O
F
Ne
Lower energy

with *s-p* mixing

- For some diatomic molecules in this period (***for elements with Z ≤ 7***), the two σ MOs (σ_{2s} and σ_{2p}) are of such similar energy and shape that they themselves mix to form modified σ orbitals (→ *s-p* mixing).
- The modified σ_{2s} (with some σ_{2p} mixed in) goes down in energy, and the modified σ_{2p} (with some σ_{2s} mixed in) goes up in energy, producing a different ordering of energy levels. **As a result, σ_{2p} is pushed up in energy above the π_{2p} orbitals.**

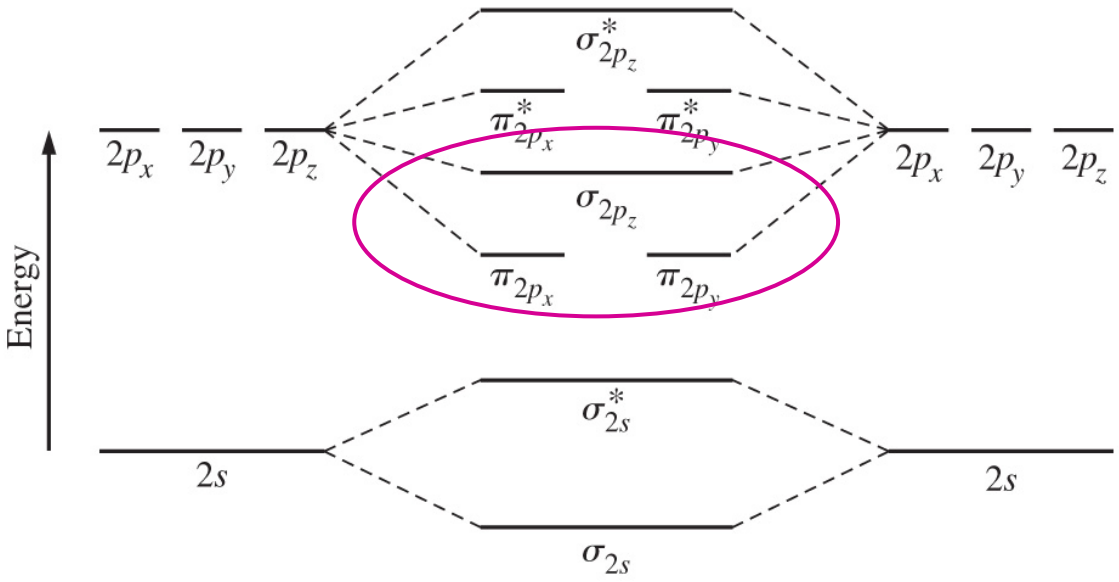
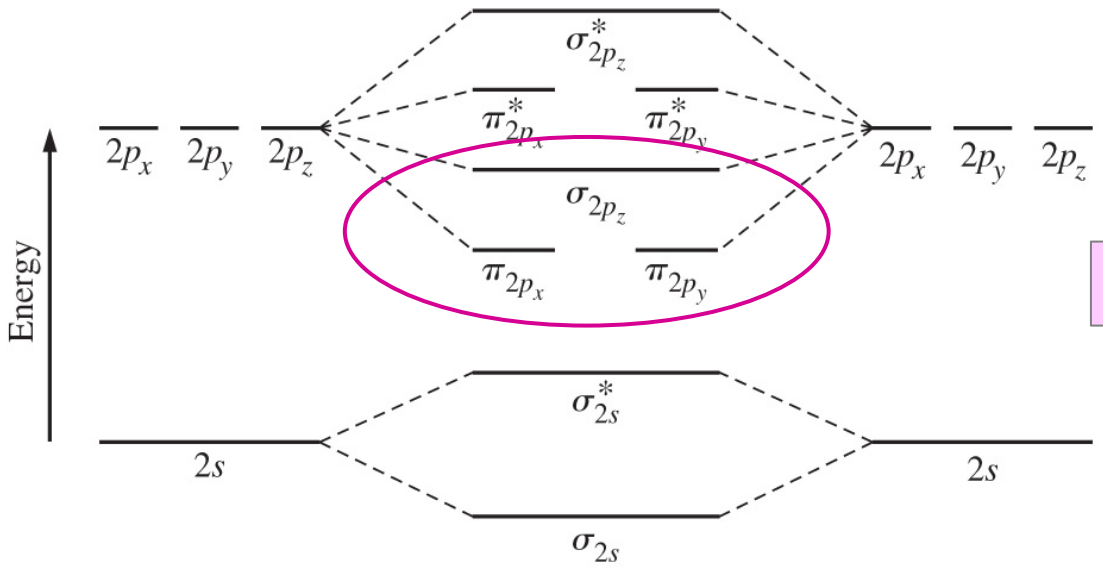


Fig. 11-25b)
MO energy level diagram for diatomic molecules of 2nd period elements **(with *s-p* mixing)**

Z ≤ 7

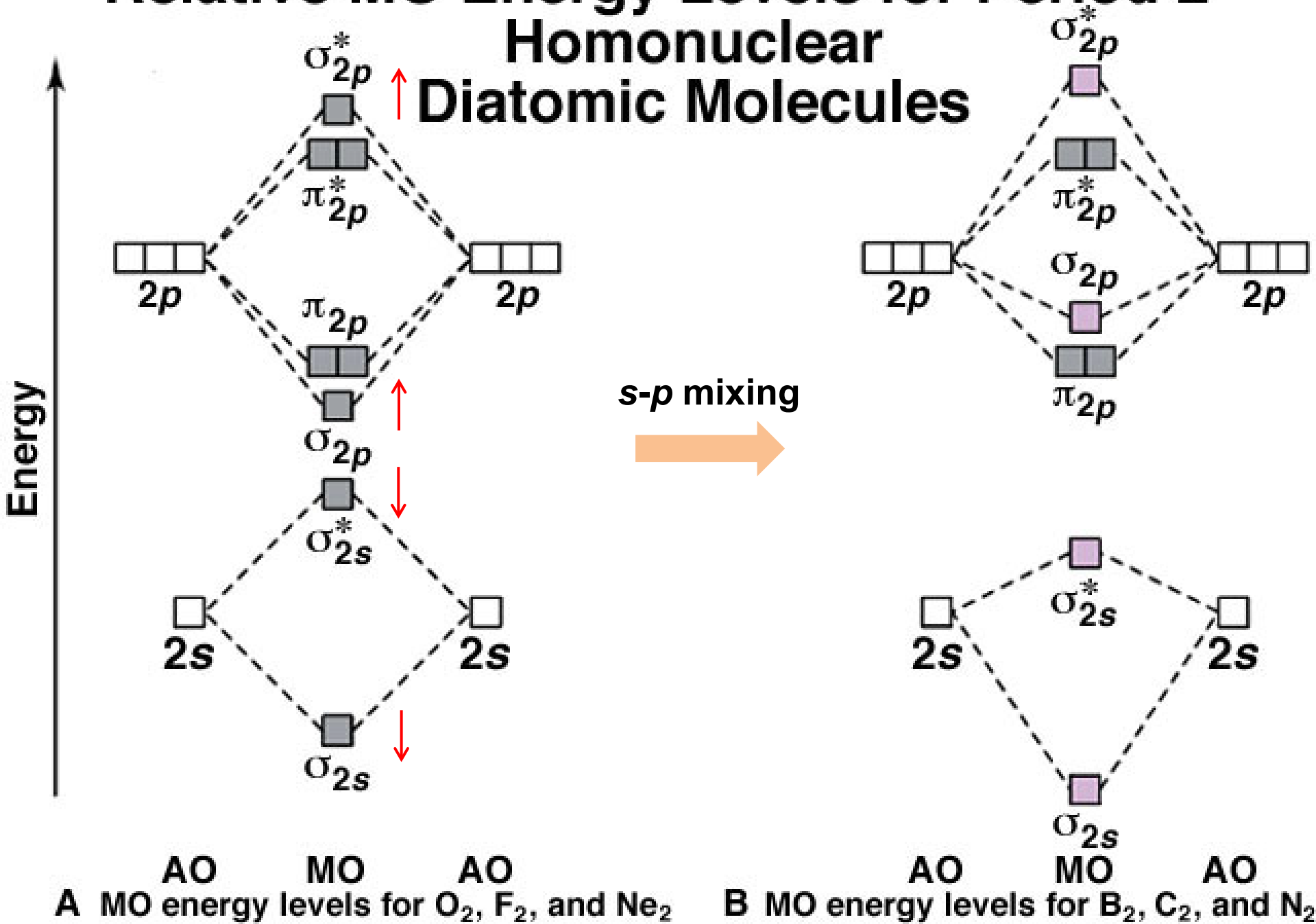


with *s*-*p* mixing

- For elements of **Z ≤ 7 (Li to N)**, the energy level of 2*s* is not so distant from 2*p*. Interactions between 2*s* and 2*p* become possible.
- These interactions affect the relative energies of σ_{2p} and π_{2p}. The σ_{2p} is pushed up in energy above the π_{2p} orbitals.
- This makes σ_{2p} of higher energy than π_{2p}.

Martin S. Silberberg, *Chemistry: The Molecular Nature of Matter and Change*, 2nd Edition. Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Relative MO Energy Levels for Period 2 Homonuclear Diatomic Molecules

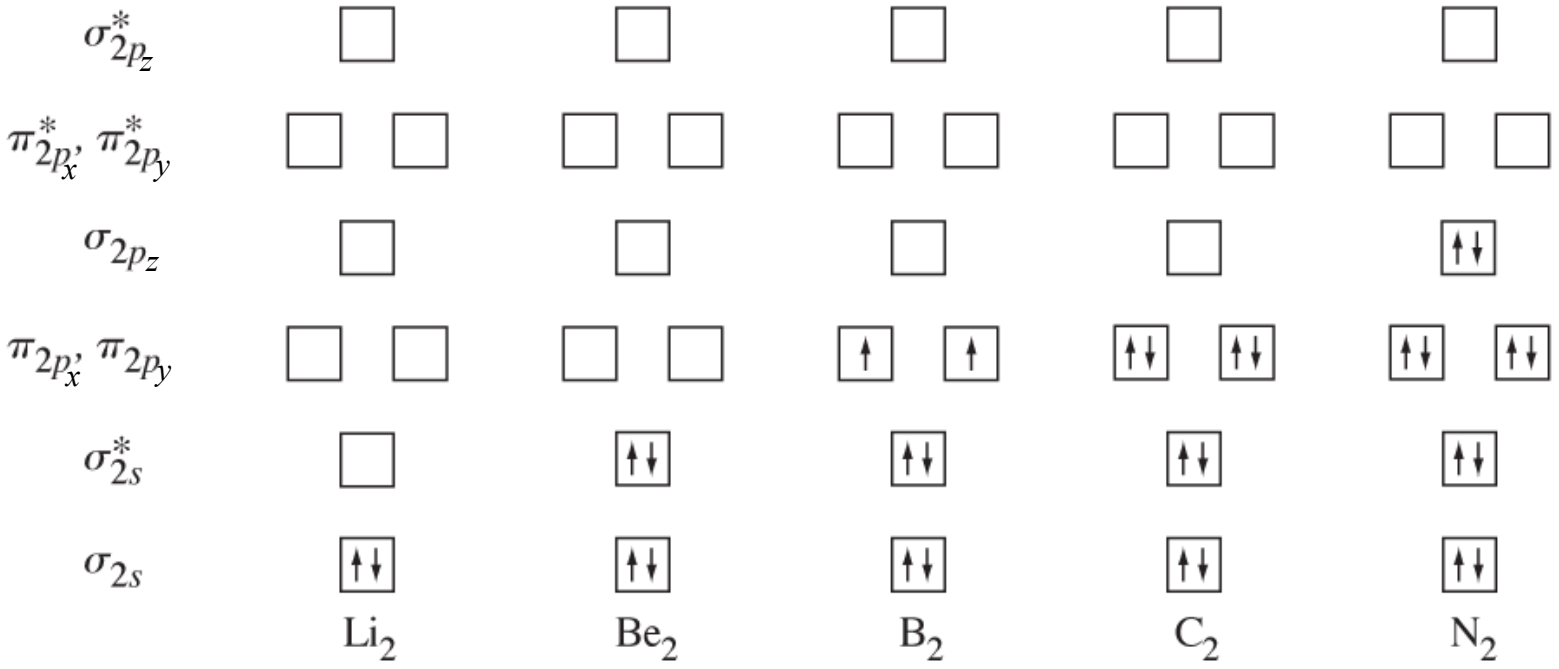


A MO energy levels for O_2, F_2, and Ne_2

B MO energy levels for B_2, C_2, and N_2

Molecular Orbital Occupancy Diagrams for the Homonuculear Diatomic Molecules of the Second-Period Elements from Lithium to Nitrogen (**Z ≤ 7**):

with *s*-*p* mixing

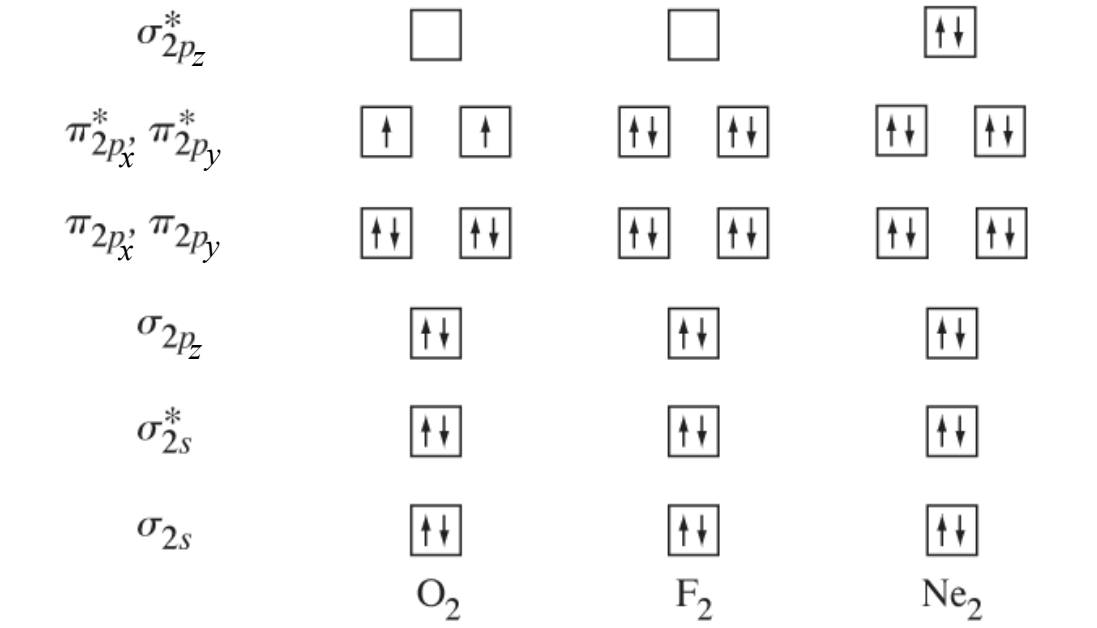


	Li_2	Be_2	B_2	C_2	N_2
Bond order	1	0	1	2	3
Magnetism	Diamagnetic	–	Paramagnetic	Diamagnetic	Diamagnetic

Not exist

Molecular Orbital Occupancy Diagrams for the Homonuculear Diatomic Molecules of the Second-Period Elements from Oxygen to Neon (**Z ≥ 8**):

without *s-p* mixing



	O_2	F_2	Ne_2
Bond order	2	1	0
Magnetism	Paramagnetic	Diamagnetic	–

Not exist

- The ***two unpaired electrons*** in O_2 explains its ***paramagnetic*** property.

Lewis theory, VSEPR theory and VB theory provide satisfactory explanations and predictions to molecular structure in most cases, but still fail to explain some aspects in bonding.

E.g.) ***Why are some molecules coloured?***

Why does O_2 interact with a magnetic field?

Information about **Magnetic Properties**:

- Two common types of magnetic behaviour:

(a) Diamagnetic – no unpaired electron

Diamagnetic substances are weakly repelled by external magnetic fields. When two spins are opposing, the magnetic fields cancel each other.

(b) Paramagnetic – with unpaired electron(s)

Paramagnetic substances are attracted by external magnetic fields. When spins are unpaired, the magnetic fields do not cancel.





Class Practice 1

Petrucci Example 11-6

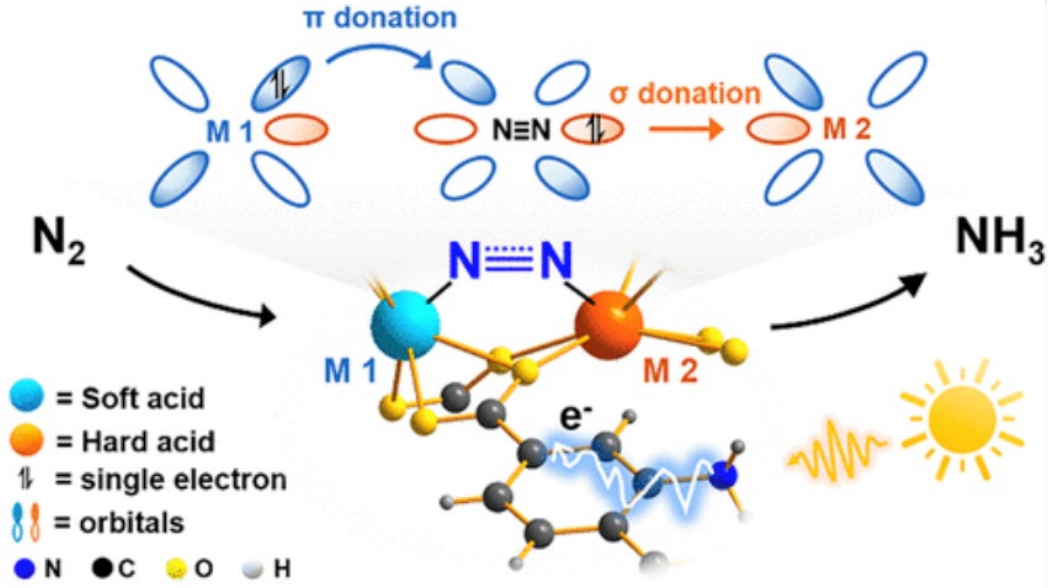
Represent bonding in O_2^+ with a molecular orbital diagram, and determine the bond order in this ion. Is it diamagnetic or paramagnetic?



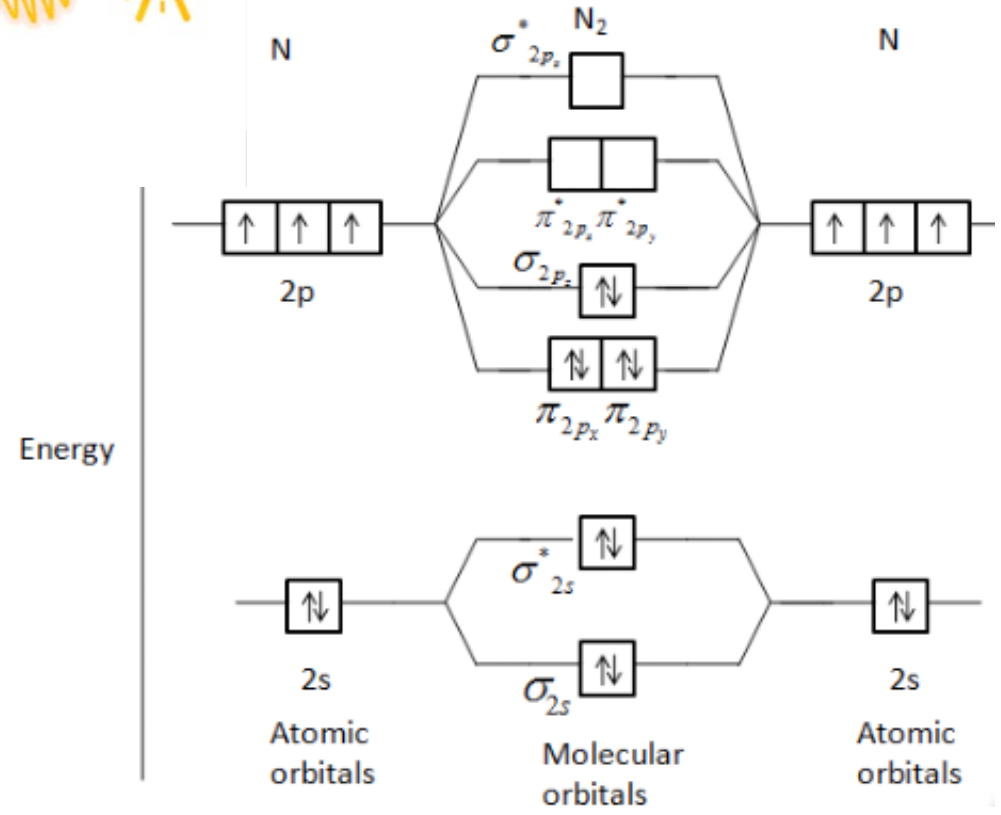
Class Practice 2

Represent bonding in B_2^{2-} with a molecular orbital diagram, and determine the bond order in this ion.

Nitrogen fixation



ACS Catal. **2021**



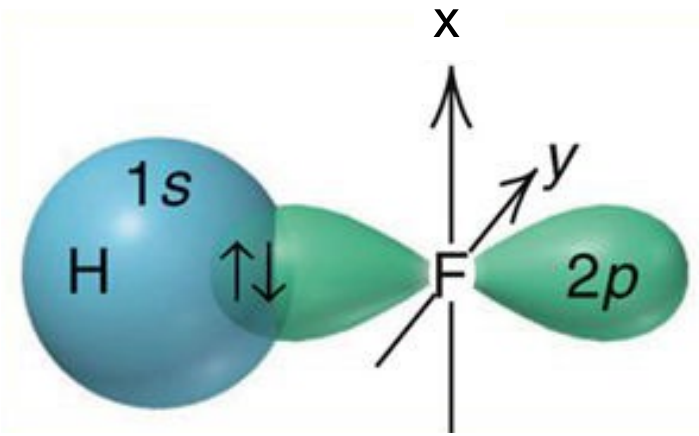
Heteronuclear Diatomic Molecules

- Based on the same ideas developed for homonuclear diatomic species.

(E.g. 1) Consider the molecule of hydrogen fluoride, HF

H $1s^1$

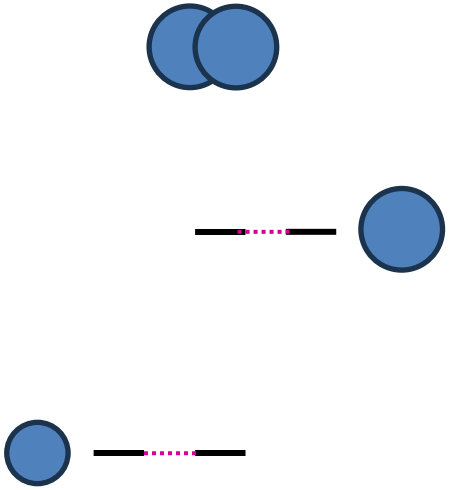
F $1s^2 2s^2 2p_x^2 2p_y^2 2p_z^1$



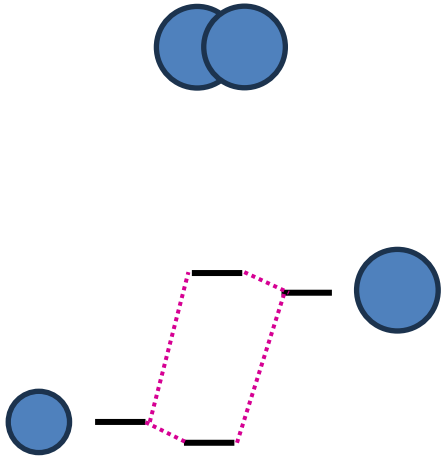
Summary of the MO diagram for the diatomic molecules of 2nd period elements:

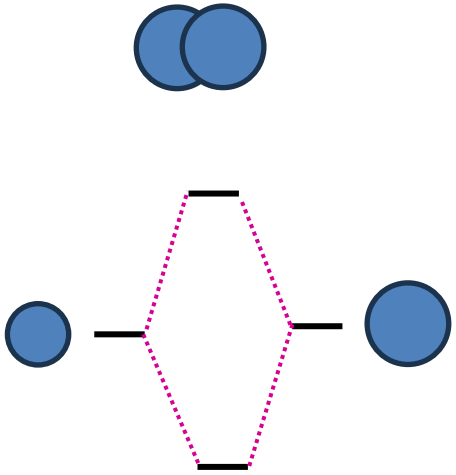
- 1s and 2s orbitals do not combine.
- σ and π orbitals do not combine.

Only AOs of similar energy and symmetry combine to form MOs.



Not similar energy



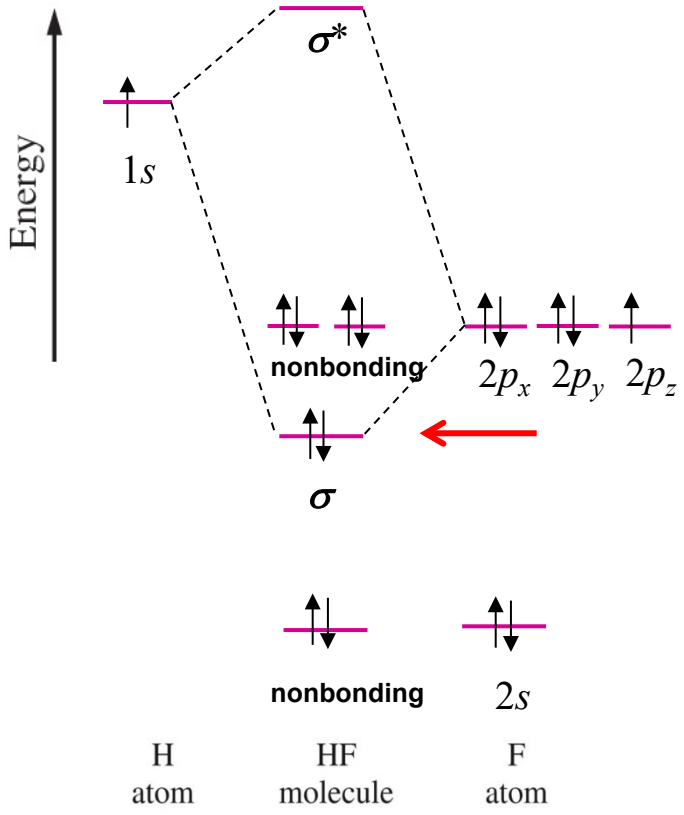


The molecular orbital diagram of HF

- The 2*s* and 2*p* orbitals of F are lower in energy than the 1*s* orbital of H because F binds its valence electrons more tightly.
- The energy gap between 2*s* of F and 1*s* of H is too large → no overlap.
- The 1*s* orbital of H overlaps with one of the 2*p* orbitals of F ($2p_z$ in this MO diagram) to produce a **σ bond**. The other two 2*p* orbitals of F ($2p_x$ and $2p_y$) have no net overlap with 1*s* of H → these two 2*p* orbitals remain as **nonbonding** in the molecule.

- The MO diagram predicts that the HF molecule should be **stable** because both electrons are lowered in energy relative to their energy in the free H and F atoms, which is the driving force for bond formation.
- The σ orbital containing the bonding electron pair shows greater electron probability close to the fluorine. The electron pair is **not** shared equally.
- This causes the fluorine atom to have a slight excess of negative charge and leaves the hydrogen atom partially positive. This is exactly the bond polarity observed for HF.

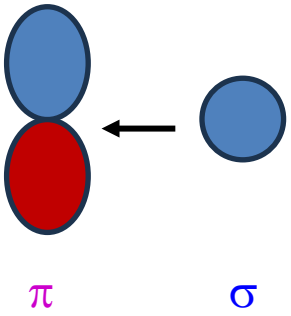
The MO diagram of HF

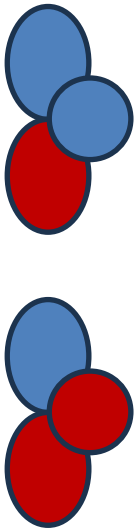


Summary of the MO diagram for the diatomic molecules of 2nd period elements:

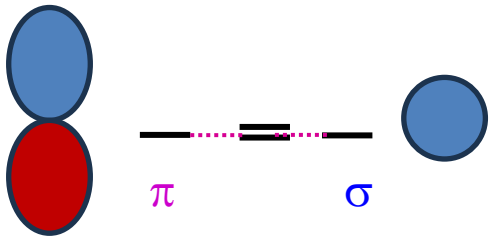
- 1s and 2s orbitals do not combine.
- σ and π orbitals do not combine.

Only AOs of similar energy and symmetry combine to form MOs.

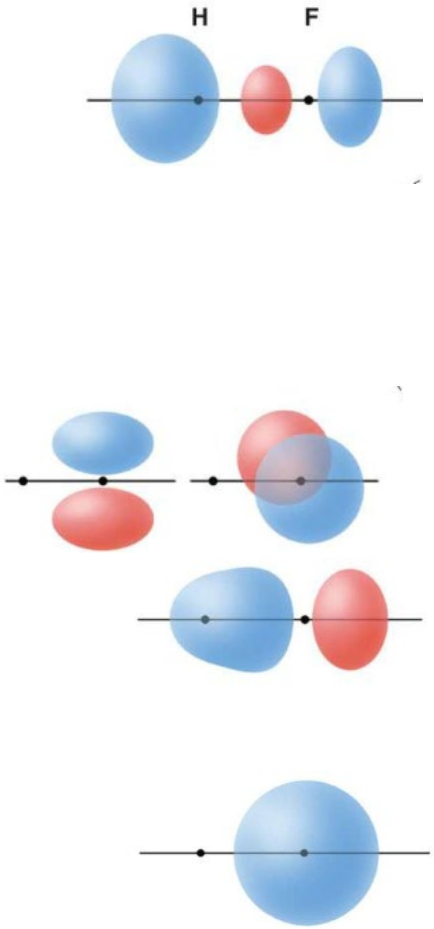




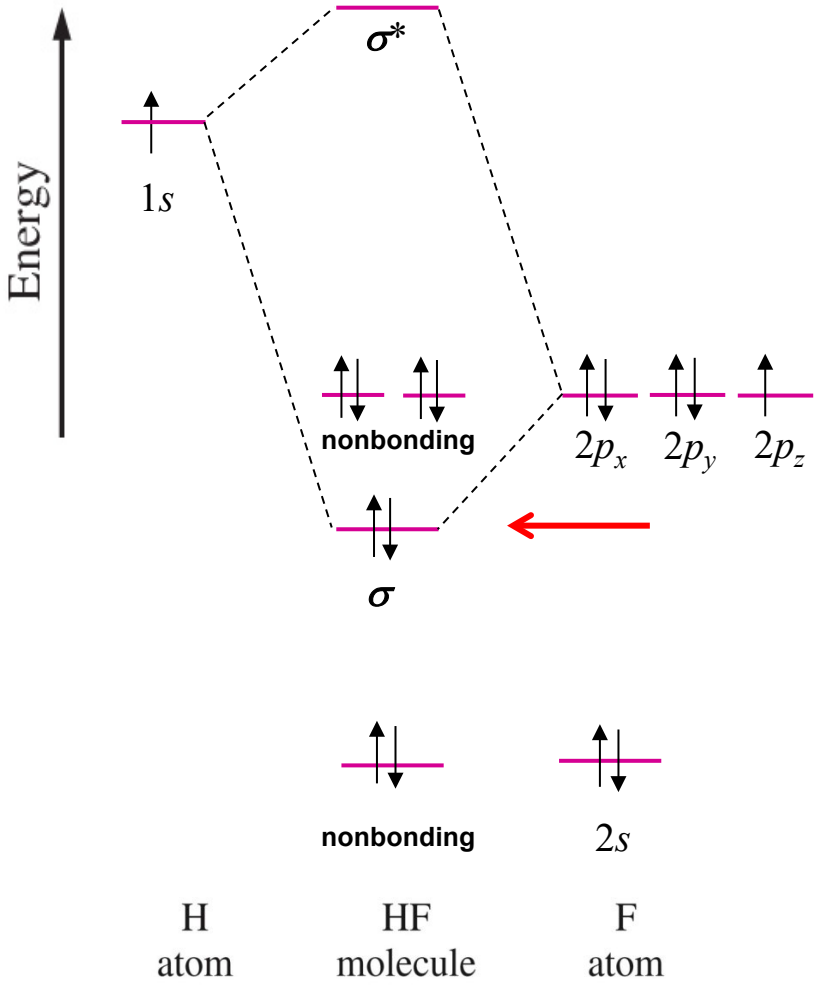
Not similar symmetry



Non bonding orbitals



The MO diagram of HF

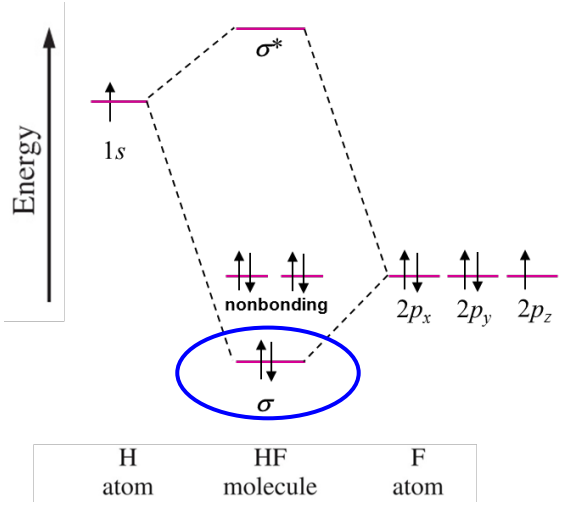


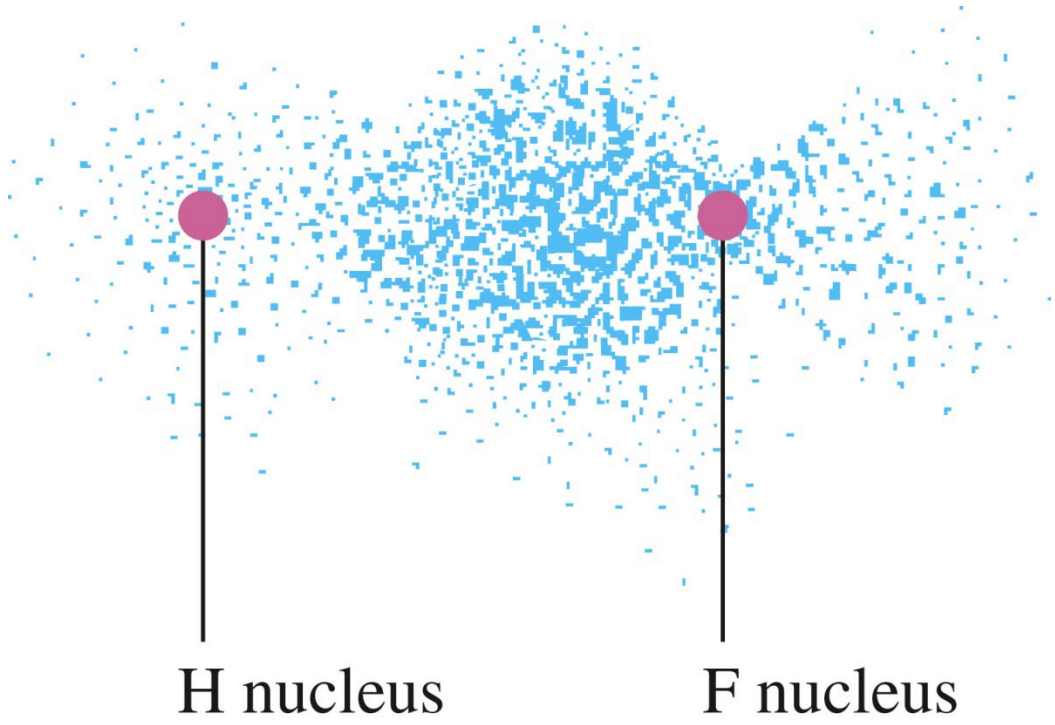
Molecular orbitals can be classified as **bonding**, **anti-bonding**, **or** **non-bonding**.

This classification is made by considering the electron density (ψ^2) in the **inter-nuclear region**.

- **Bonding** orbitals will have electron density between the nuclei;
- **Anti-bonding** orbitals will have a node between the nuclei;
- **Non-bonding** orbitals neither **add to** nor **detract from** bond formation.

The Electron Probability Distribution in the Bonding Molecular Orbital of the HF Molecule





Heteronuclear Diatomic Molecules

(E.g. 2) Consider the molecule of carbon monoxide, CO.

1. First, consider the relative placement of the 2*s* and 2*p* orbitals of C and O.

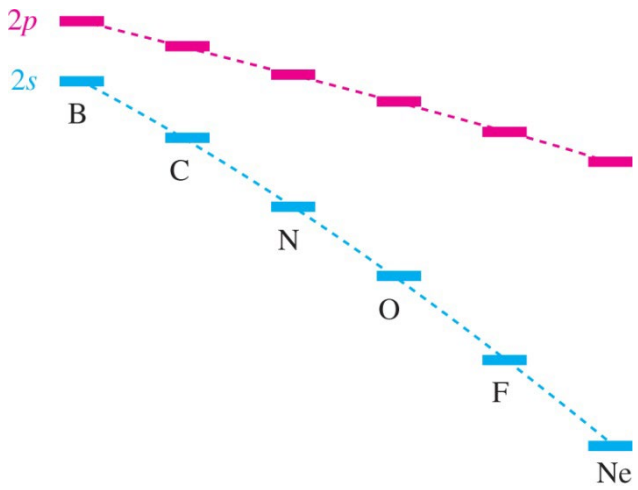


Fig 11-27) The relative energies of 2*s* and 2*p* orbitals of some second-period elements

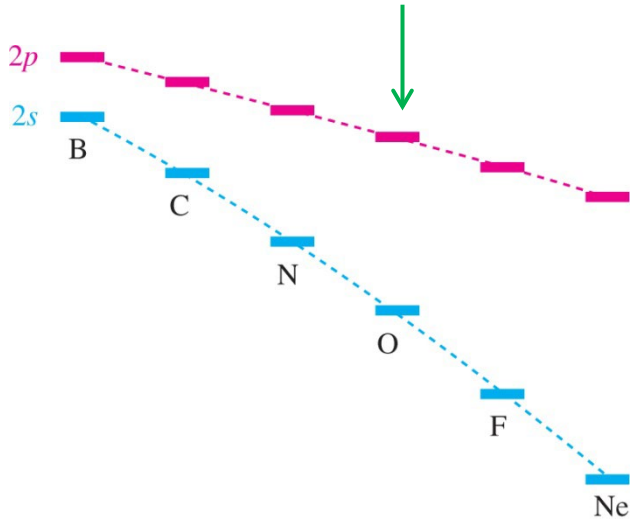
- The 2*s* and 2*p* orbitals decrease in energy from left to right across a period because of an increase in the effective nuclear charge.
- The energy separation between the 2*s* and 2*p* orbitals also increases from left to right across the period.

2. Second, decide whether or not the σ MOs of CO should be formulated by considering only the combinations of C(2*s*) with O(2*s*) and C(2*p*) with O(2*p*). *(To be continued on next slide)*

Heteronuclear Diatomic Molecules

(Con't)

2. From Fig. 11-27), in terms of energy, the 2*p* orbital of O is halfway between the 2*s* and 2*p* orbitals on C. Thus, the combination of O(2*p*) with C(2*s*) is probably as important as the combination of O(2*p*) with C(2*p*).

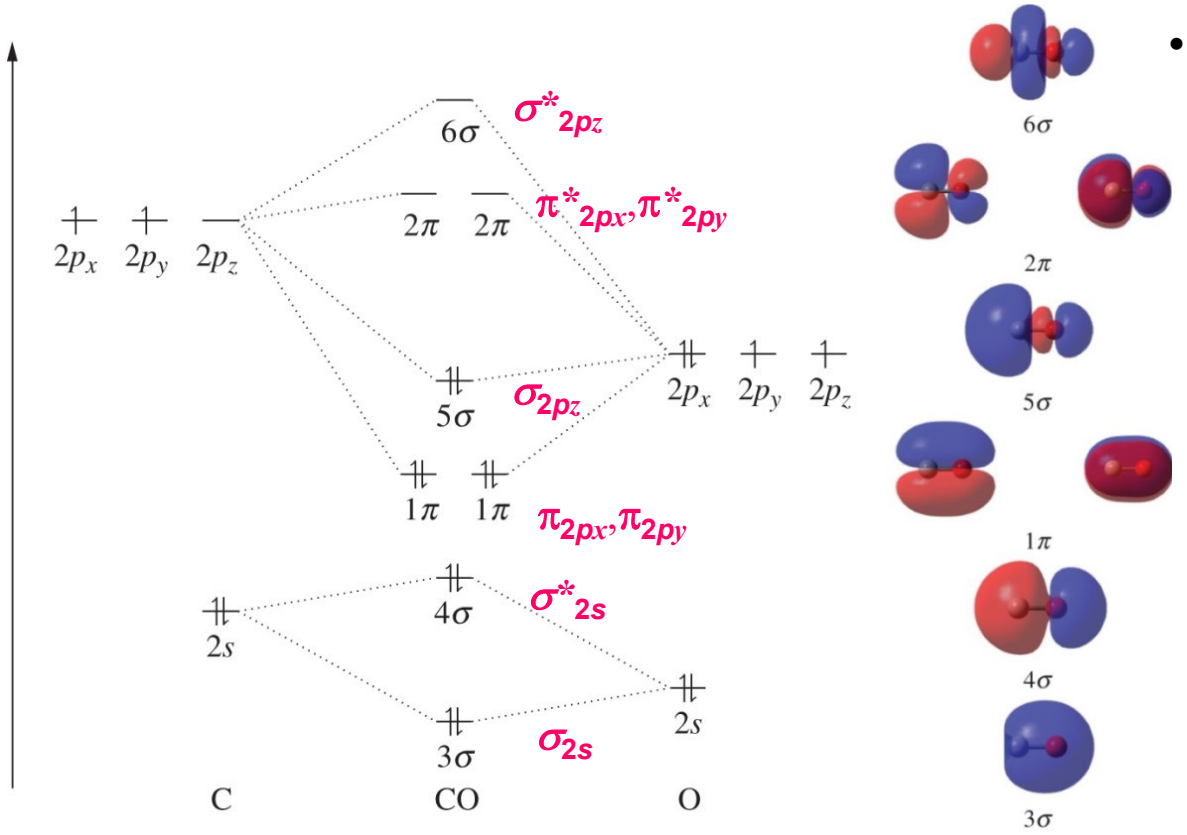


∴ it is expected that the σ orbitals of CO to be mixtures of 2*s* and 2*p* orbitals from both C and O and, consequently, the energy ordering of the MOs of CO will be similar to that shown in Fig. 11-25b.

There is s-p mixing

Fig. 11-28) The molecular orbital diagram of CO

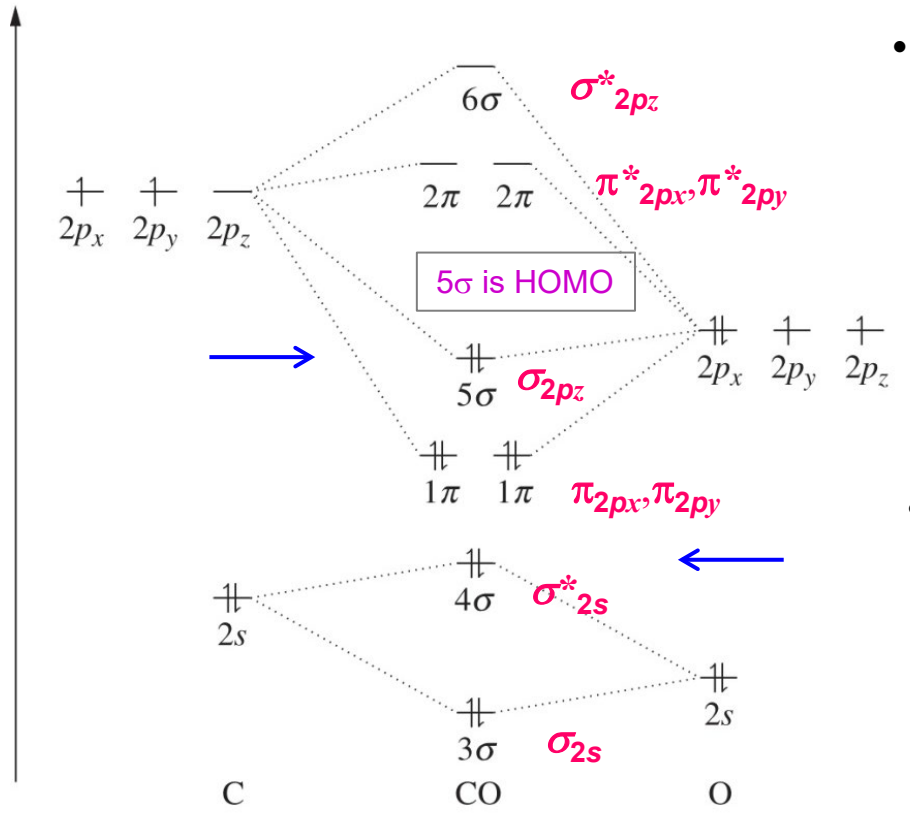
Note: (1) Contributions to the MOs from each atoms are unequal.
(2) This MO diagram is the results of quantum mechanical calculations.



- The MOs are labeled by using a simplified notation in which the symbol σ or π is preceded by an integer to indicate the relative energy ordering of the orbitals of each type. Thus the σ orbital of lowest energy is labeled 1σ, the next lowest 2σ, and so on. Similarly, the π orbitals of lowest energy is 1π, the next lowest is 2π, and so on.

- The 1π orbitals (bonding) and 2π orbitals (antibonding) are mixtures of $2p$ orbitals from C and O. The 3σ (bonding), 4σ (nonbonding), 5σ (nonbonding), and 6σ (antibonding) orbitals are **mixtures** of $2s$ and $2p$ orbitals from both C and O. The 1σ and 2σ orbitals which are mixtures of the $1s$ orbitals from C and O are not shown.

(Con't) The molecular orbital diagram of CO



- The configuration of CO is therefore

$$3\sigma^2 4\sigma^{\,2} 1\pi^4 5\sigma^{\,2}$$

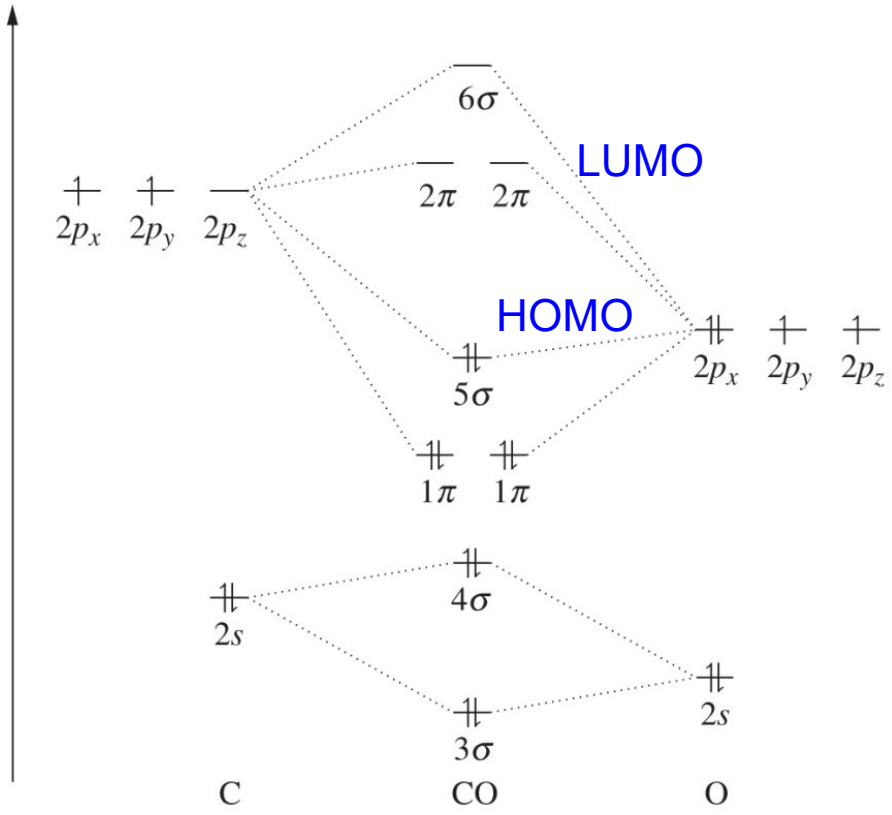
Or $\sigma_{2s}^2\ \sigma_{2s}^{*2}\ (\pi_{2p_x},\pi_{2py})^4\ \sigma_{2p_z}^2$

with the 5σ orbital the highest occupied molecular orbital (HOMO).

- The 1π orbitals are bonding whereas 2π are antibonding. This is established by recalling that the π orbitals are mixtures of $2p$ orbitals from C and O, and observing that the 1π and 2π are, respectively, lower and higher in energy than the $2p$ orbitals from C and O.

- As for σ orbitals, all of them are mixtures of $2s$ and $2p$ from both C and O. Because the 3σ and 6σ are, respectively, lower and higher in energy than the $2s$ and $2p$ of C and O, we may tentatively classify 3σ as bonding and 6σ as antibonding. We observe that 4σ is approximately halfway between $2s$ and $2p$ on O, whereas 5σ is approximately halfway between $2s$ and $2p$ on C. It is reasonable to classify them as nonbonding.

The molecular orbital diagram of CO



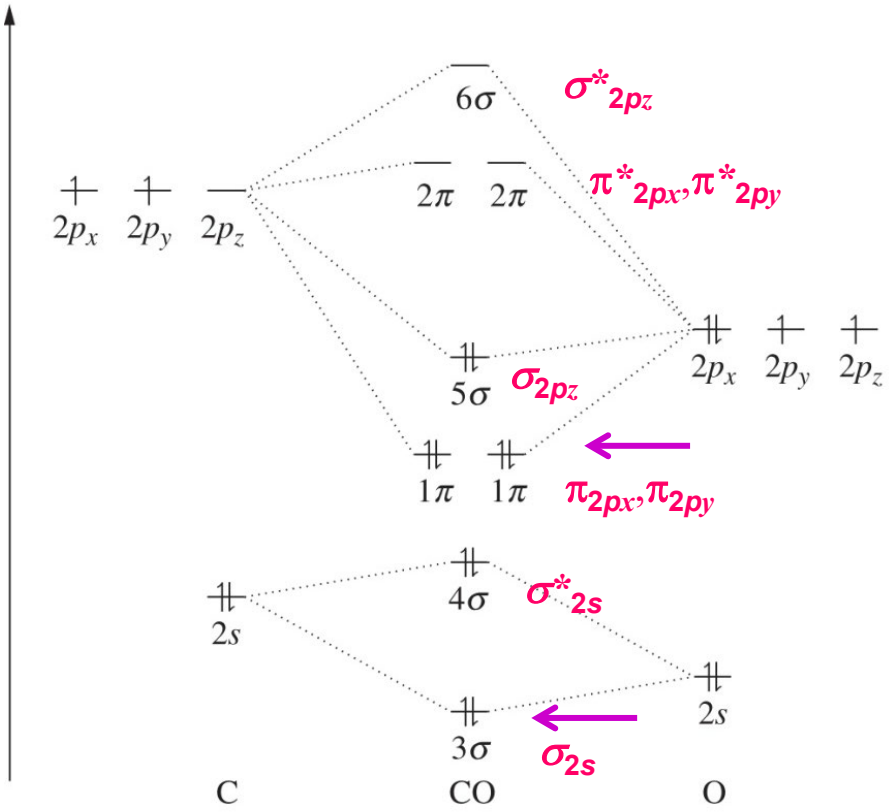
Frontier Molecular Orbitals – HOMO and LUMO

Highest **O**ccupied **M**olecular **O**rbital

Lowest **U**noccupied **M**olecular **O**rbital

The orbitals most likely to be involved in chemical reactivity

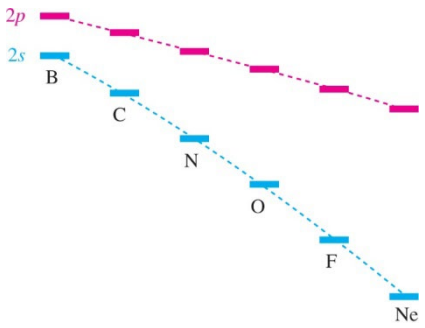
(Con't) The molecular orbital diagram of CO



- The bonding in CO comes from the two electrons in the 3σ orbital and the two electron pairs in the degenerate 1π orbitals, producing an effective **bond order of 3**.

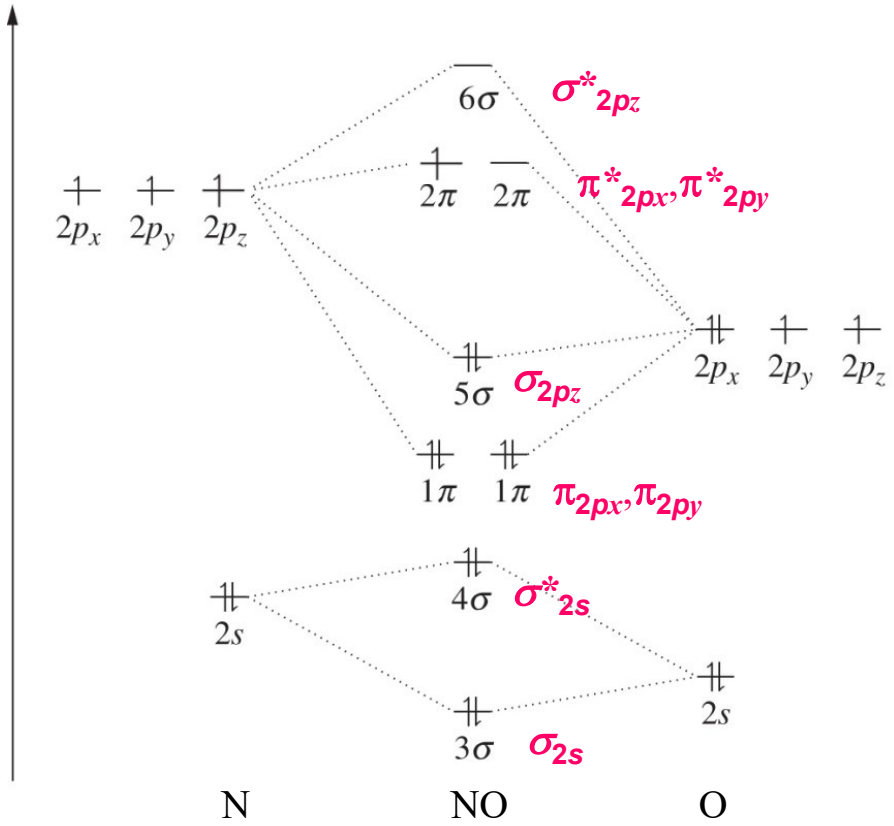
This diagram is appropriate for many heteronuclear diatomic molecules of second-period elements.

Heteronuclear Diatomic Molecules



(E.g. 3) Consider the molecule of nitric oxide (NO)

- The MO diagram of NO is similar to that of CO,



What is the bond order and magnetic properties of NO?

$3\sigma^2 4\sigma^2 1\pi^4 5\sigma^2 2\pi^1$

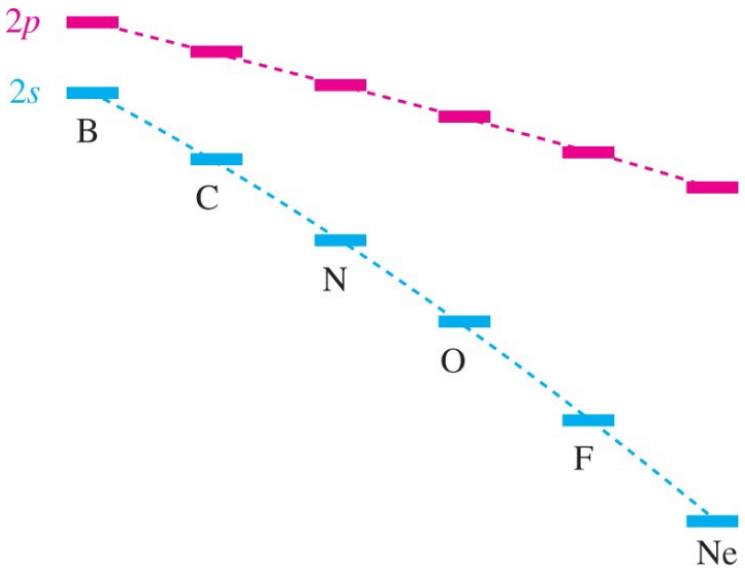
Or $\sigma_{2s}^2\ \sigma_{2s}^{*2}\ (\pi_{2p_x}, \pi_{2py})^4\ \sigma_{2p_z}^2\ (\pi_{2p_x}^*, \pi_{2py}^*)^1$

Bond order of NO = (8 – 3) / 2
= 2.5

NO is paramagnetic because it has an unpaired electron.

Class Practice 3

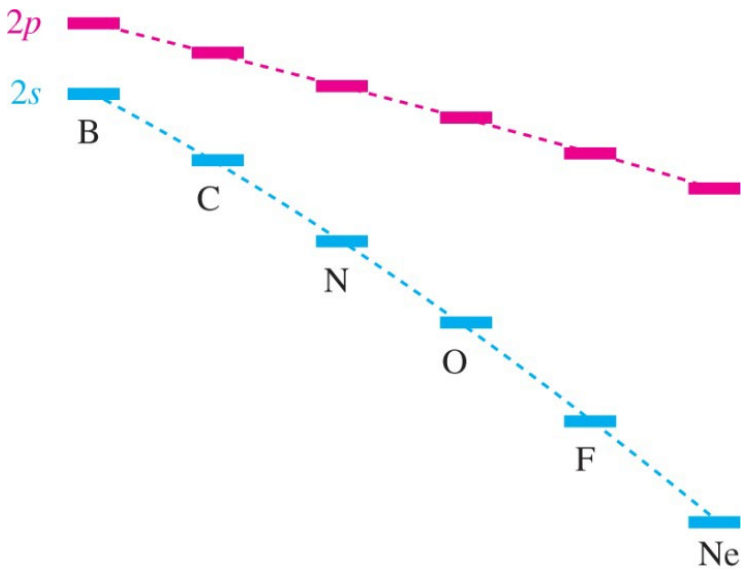
Represent bonding in BO with a molecular orbital diagram, and determine its bond order. Is it diamagnetic or paramagnetic?



Class Practice 4

EXAMPLE 11-7 Writing Electron Configurations for Heteronuclear Diatomic Species

Write the ground state electron configuration for the cyanide ion, CN^-, and determine the bond order for this ion. Clearly state any assumptions.





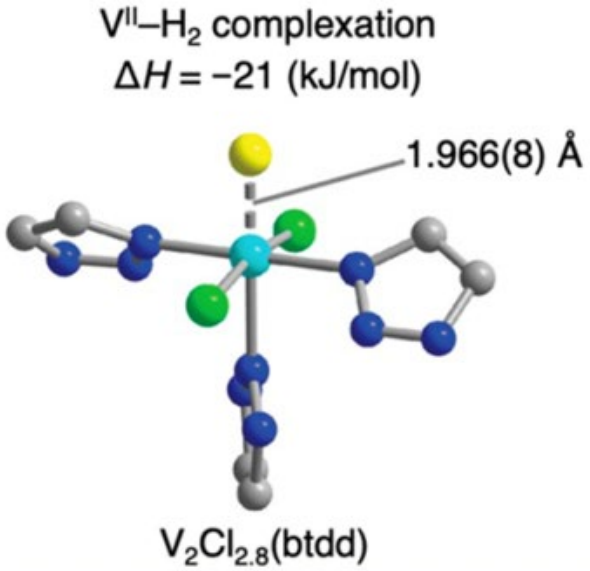
Class Practice 5

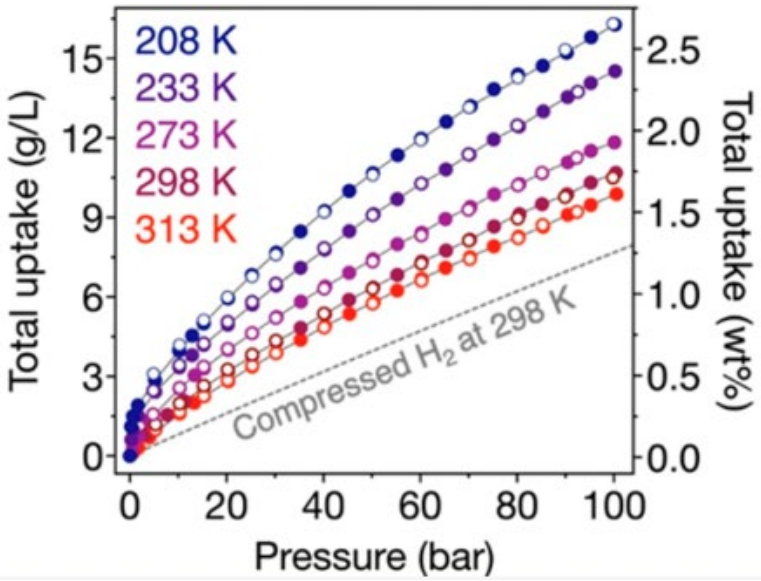
Write the electron configuration for each of the following, and determine the bond order in each case.

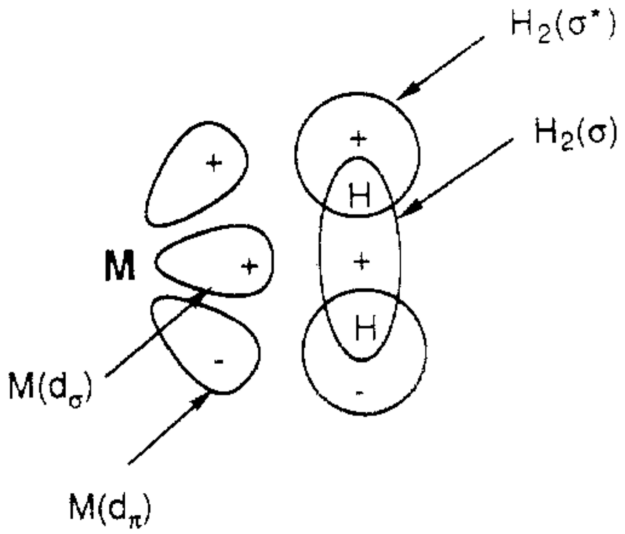
(a) CN^+

(b) BN

Hydrogen storage



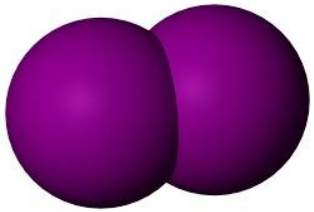




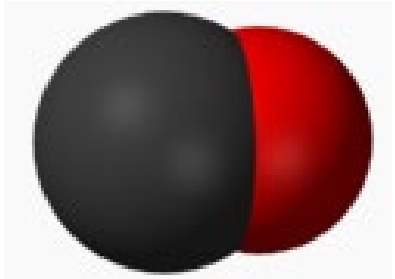
J. Am. Chem. Soc. **2021**

Molecular Orbital Theory

Diatomics

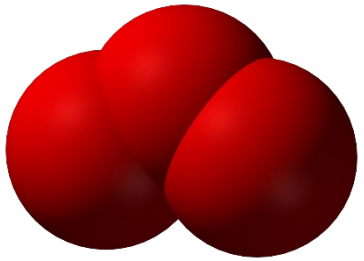




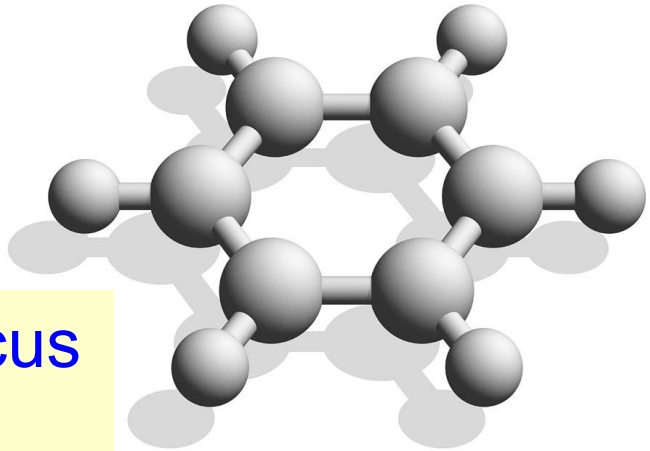


Group orbitals,
Group theory

Polyatomics





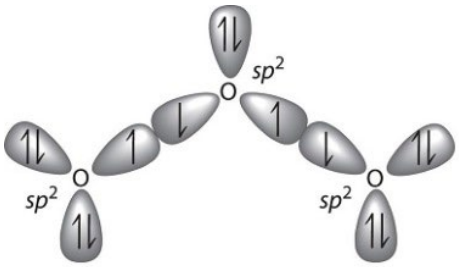


Here, let us focus on the π–MOs.

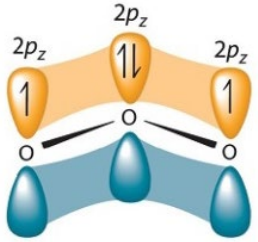
Delocalized Electrons (1): Bonding in Ozone

- Consider the **ozone** (O_3) molecule, the resonance hybrid based on the following two contributing structures can be written as the single structure shown in the following figures (a & b).





(a) σ bond framework



(b) Delocalized π molecular orbital

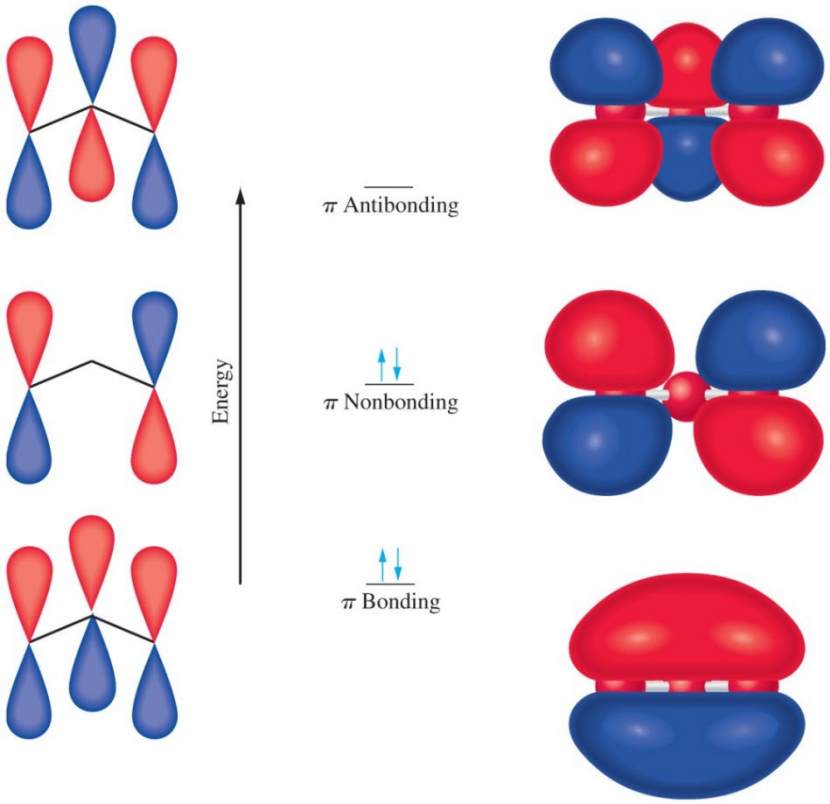
- With VSEPR theory, the ozone molecule has a trigonal planar e-group geometry (actual bond angle is 117°). The hybridization scheme for the central O atom is ***sp²*** and the terminal O atoms have the same hybridization scheme too (i.e. each O atom uses the orbital set ***sp² + p***).

- Of the 18 valence electrons in O_3, assign 14 to the sp^2 hybrid orbitals of the σ-bond framework (see figure a). Four of these are bonding electrons and ten are lone-pair electrons (see figure b).

Delocalized Electrons (1): Bonding in Ozone

- The three unhybridized 2*p* orbitals combine to form *three* molecular orbitals of the π type (see figure below). One of these orbitals is bonding, and the second is antibonding. The third is a *nonbonding molecular orbital*. (A nonbonding molecular orbital has the same energy as the atomic orbitals from which it is formed, and it neither adds to nor detracts from bond formation.)

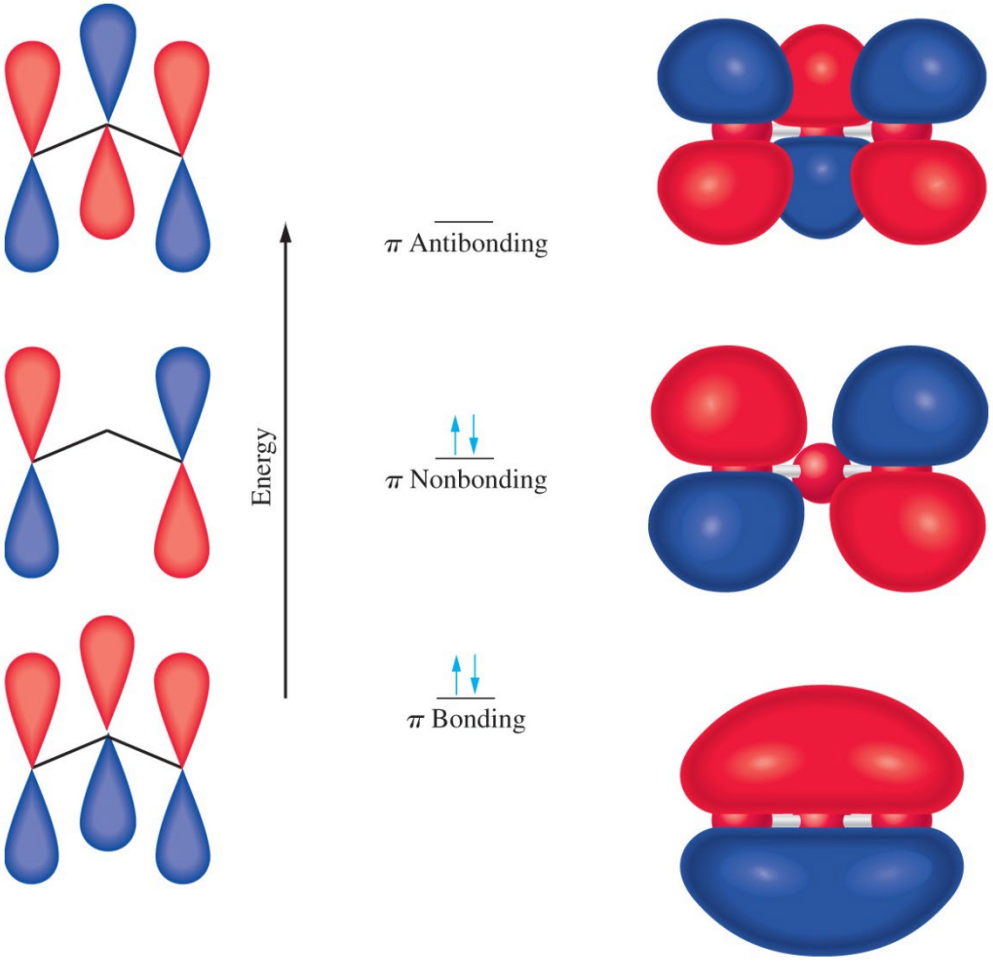
π bonding orbitals of the ozone molecule O_3



- The remaining ***four*** valence electrons are assigned to the **π MOs**. Two go into the bonding orbital and two into the nonbonding orbital. The antibonding orbital remains empty.

- The bond order associated with the π MOs is (2 – 0) / 2 = 1. This π bond is distributed between the two O–O bonds and amounts to one-half of a π bond for each. Together with the σ bonds, the **total bond order for the O–O bond in O_3 is 1.5**.

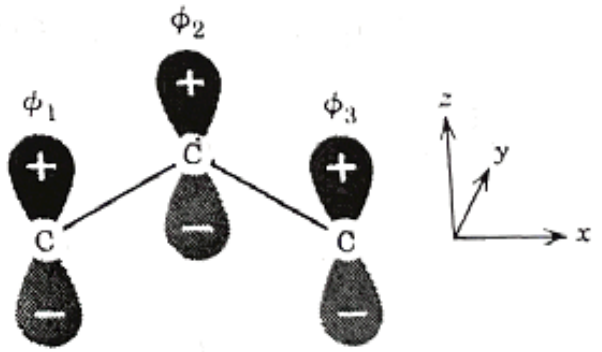
π bonding orbitals of the ozone molecule O_3



The π antibonding molecular orbital has all the three 2*p* orbitals **out of phase**.

The π nonbonding molecular orbital is at the **same energy** as the original 2*p* orbitals on the O atoms.

The π molecular orbital has all the 2*p* orbitals **in phase**.



C2v character table

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v'(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	−1	−1	R_z	xy
B_1	1	−1	1	−1	x, R_y	xz
B_2	1	−1	−1	1	y, R_x	yz

Symmetry consideration

C_{2v}	E	C_2	σ_{yz}	σ_{xy}
Γ_π	3	−1	1	−3

$$\Gamma_\pi = A_2 + 2B_1$$

Orbital outcomes

$$\psi_A^{(1)} = \tfrac{1}{2}(\sigma_1 + \sqrt{2}\sigma_2 + \sigma_3)$$

$$\psi_A^{(2)} = \tfrac{1}{2}(\sigma_1 - \sqrt{2}\sigma_2 + \sigma_3)$$

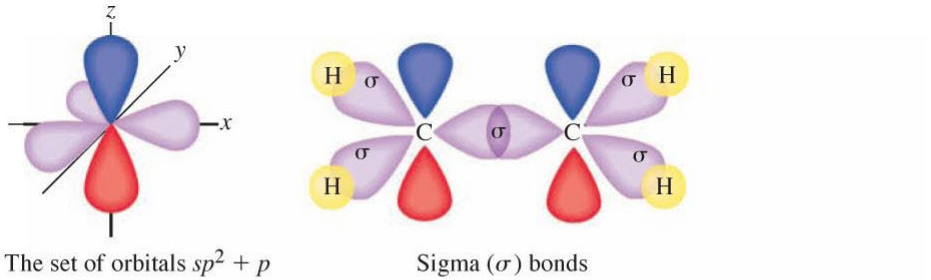
$$\psi_B = \frac{1}{\sqrt{2}}(\sigma_1 - \sigma_3)$$

Energy consideration

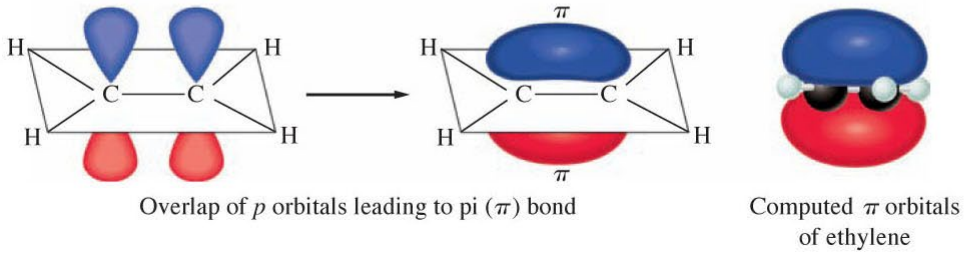
$$\begin{vmatrix} H_{11} - E & H_{12} - ES_{12} \\ H_{21} - ES_{21} & H_{22} - E \end{vmatrix} = 0$$

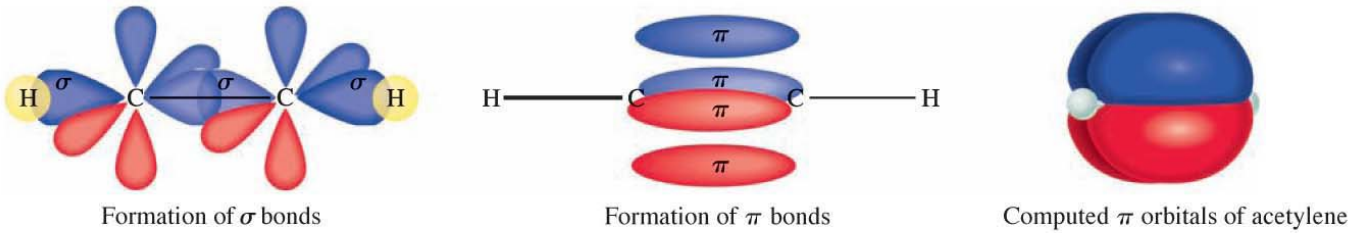
Delocalized Electrons (2): Bonding in Benzene

- Recall what you have learnt about ***localized*** π bonds, such as those in ethylene ($H_2C{=}CH_2$) and acetylene ($HC{\equiv}CH$).



◀ FIGURE 11-14
Sigma (σ) and pi (π) bonding in C_2H_4

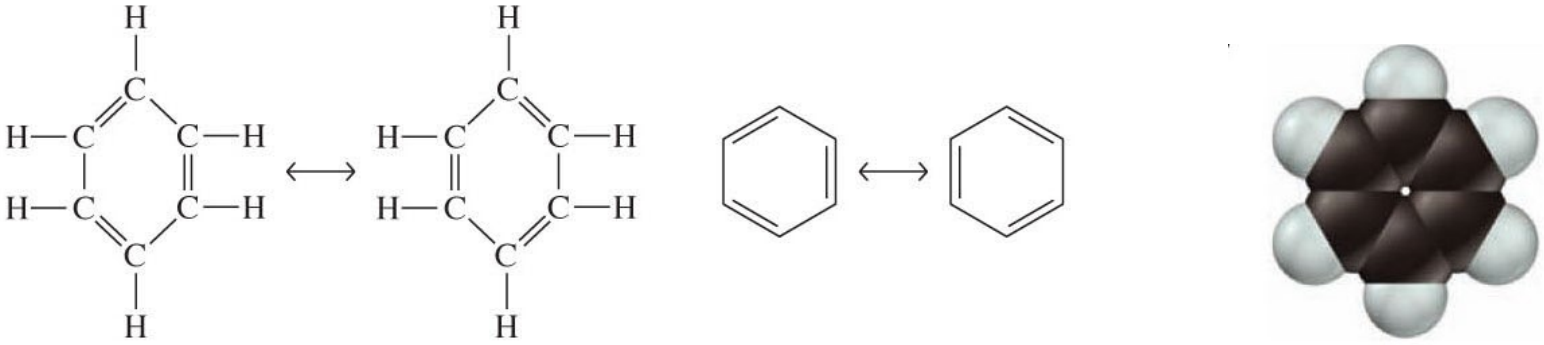




▲ FIGURE 11-16
σ and π bonding in C_2H_2

Delocalized Electrons (2): Bonding in Benzene

Consider the **benzene** (C_6H_6) molecule,



(a) (b) (c)

Resonance in benzene and the Kekulé structures

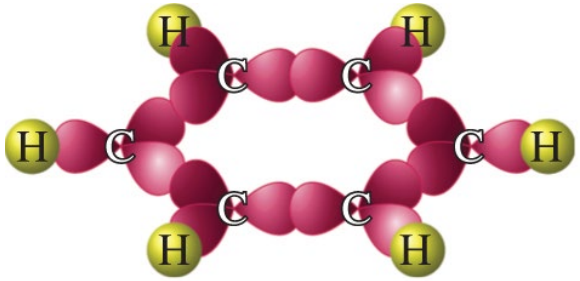
(a)(b) The two possible Kekulé structures are contributing structures to a resonance hybrid.

(c) A space-filling model.

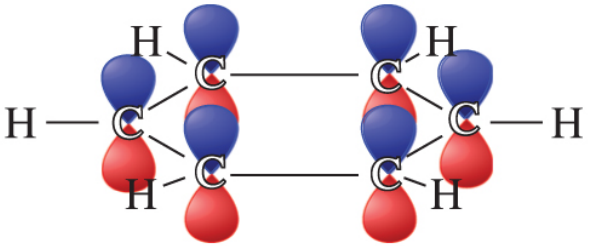
Bonding in benzene by the Valence Bond Theory

- a σ-bond framework for the observed planar structure can be constructed with 120° bond angles by using sp^2 hybridization at each carbon atom. End-to-end overlap of the sp^2 orbitals produces σ bonds.

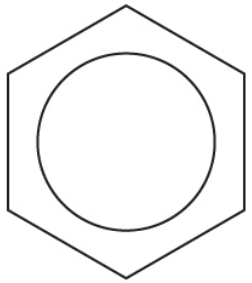
- The six remaining 2*p* orbitals are used to construct the ***delocalized*** π bonds. Note that the π bonds are **not** localized between specific carbon atoms but are spread out around the six-membered ring. In figure c), the circle represents this delocalized π bonding.



(a) σ bond framework

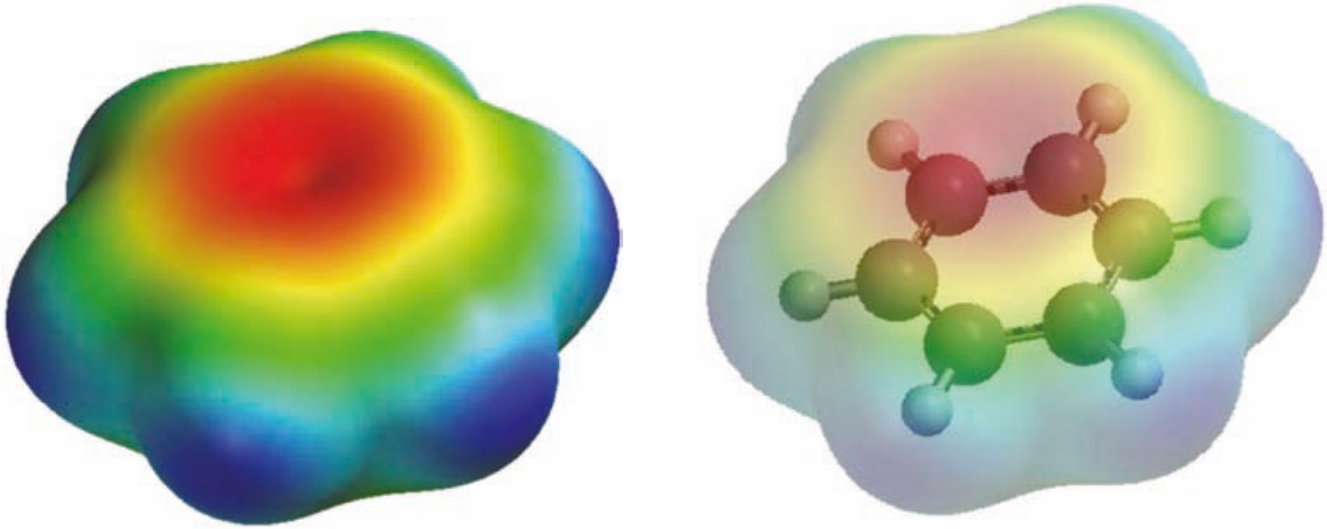


(b) Carbon 2*p* orbitals to be used in π bonding



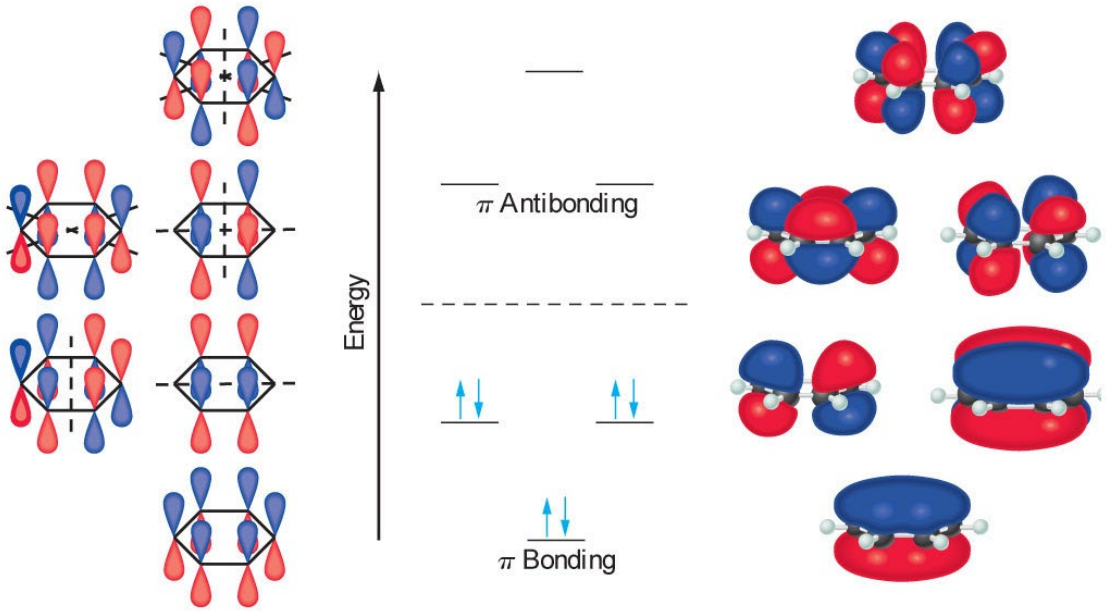
(c) Symbolic representation

Electrostatic potential maps of benzene (one solid and one transparent) showing the negative charge density caused by the π molecular orbitals (delocalized molecular orbitals) of benzene.



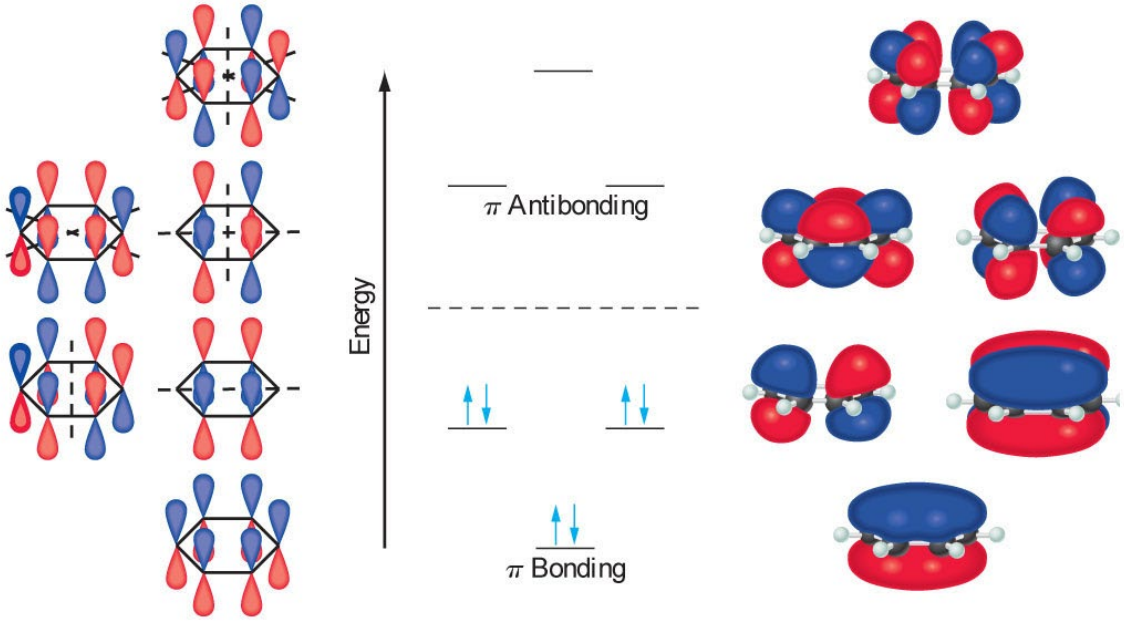
Bonding in benzene by the Molecular Orbital Theory

- To best understand the **delocalized** π bonding, MO theory is used.



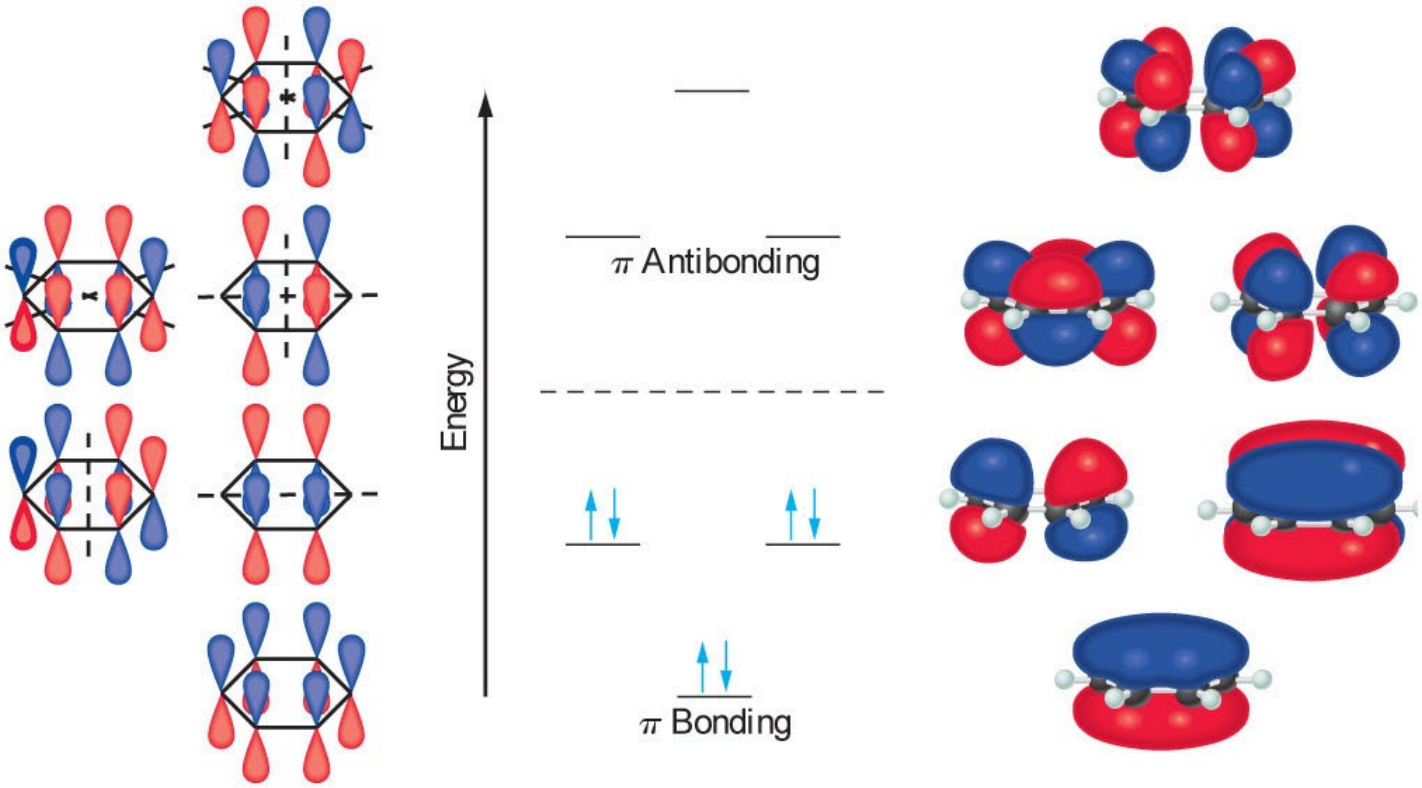
▲ FIGURE 11-30
π molecular orbital diagram for C_6H_6

- Six 2*p* AOs of the C atoms combine to form **six MOs of the π type**. Three of these π-type MOs are bonding, and three are antibonding.
- Recall that as the E of orbitals ↑, the no. of nodes ↑, so the **lowest π MO** will not possess a node. **All six 2*p* orbitals are in phase** as indicated by the fact that all the blue lobes are on one side of the σ framework.
- The **next two π bonding MOs** each have one node (dashed line in fig. a) and consequently have the same energy (i.e. degenerate).



▲ FIGURE 11-30
π molecular orbital diagram for C_6H_6

- The next pair of orbitals, **the antibonding π orbitals**, have two nodes, and **the final orbital** has three nodes (see fig. a).
- The three bonding orbitals fill with six electrons (one *2p* electron from each C atom), and the three antibonding orbitals remain empty. The bond order in this π system is (6 – 0)/2 = 3. The three bonds are distributed among the six C atoms, which amounts to 3/6 = ½ (i.e. a half-bond), between each pair of C atoms.
- Together with the σ bonds, the **total bond order for each C–C bond is 1.5.**



▲ FIGURE 11-30
π molecular orbital diagram for C_6H_6

Class Practice 6

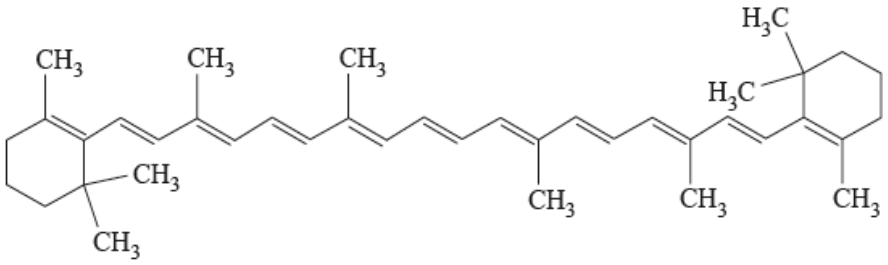
EXAMPLE 11-8 Using Delocalized Molecular Orbitals to Describe Bonding in NO_3^-

For the NO_3^- ion, one molecular orbital in the π system is bonding and one is antibonding. How many nonbonding π orbitals are there? Which of the π molecular orbitals are occupied? What is the predicted bond order for the nitrogen-to-oxygen bonds?

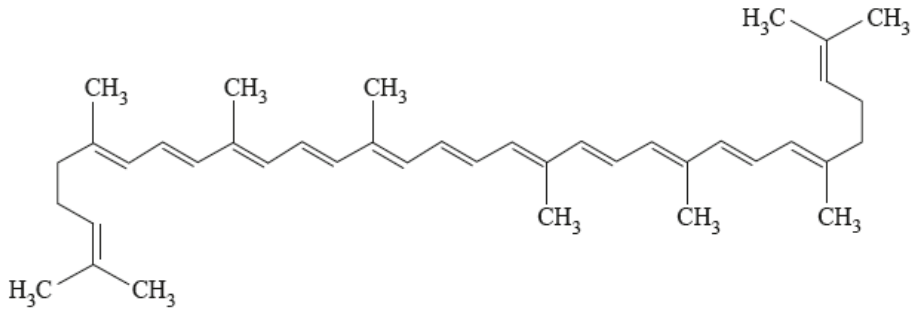
Analyze

First, we focus on electrons associated with the σ bond framework and determine hybridization schemes for the atoms. Then, we determine the number of electrons in the π system, and the number of $2p$ orbitals involved forming the π molecular orbitals. The key to solving this problem is reasoning out the number of each type of molecular orbital (bonding, antibonding, nonbonding) in the π system. The final step is to assign electrons in the π system to the appropriate molecular orbitals.

MO theory can be used to explain the colour properties of some substances. Consider the following two pigment molecules isolated from vegetables.



β-Carotene, $C_{40}H_{56}$



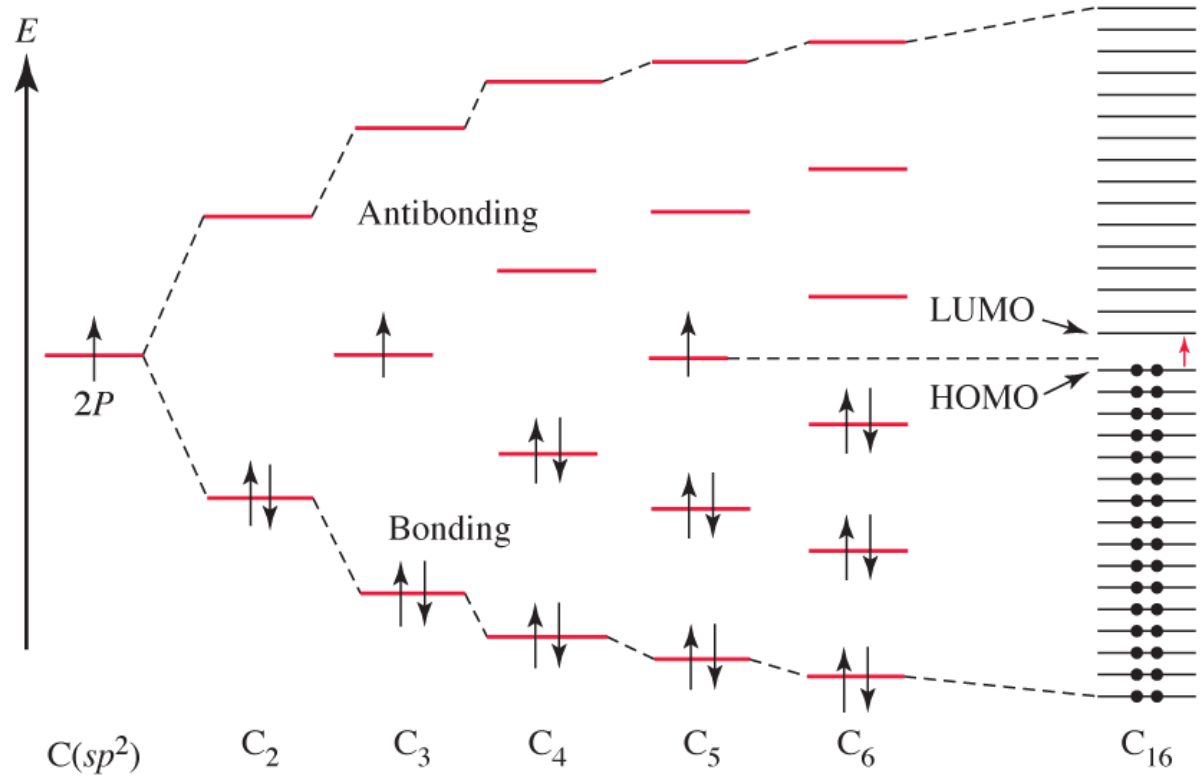
Lycopene, $C_{40}H_{58}$





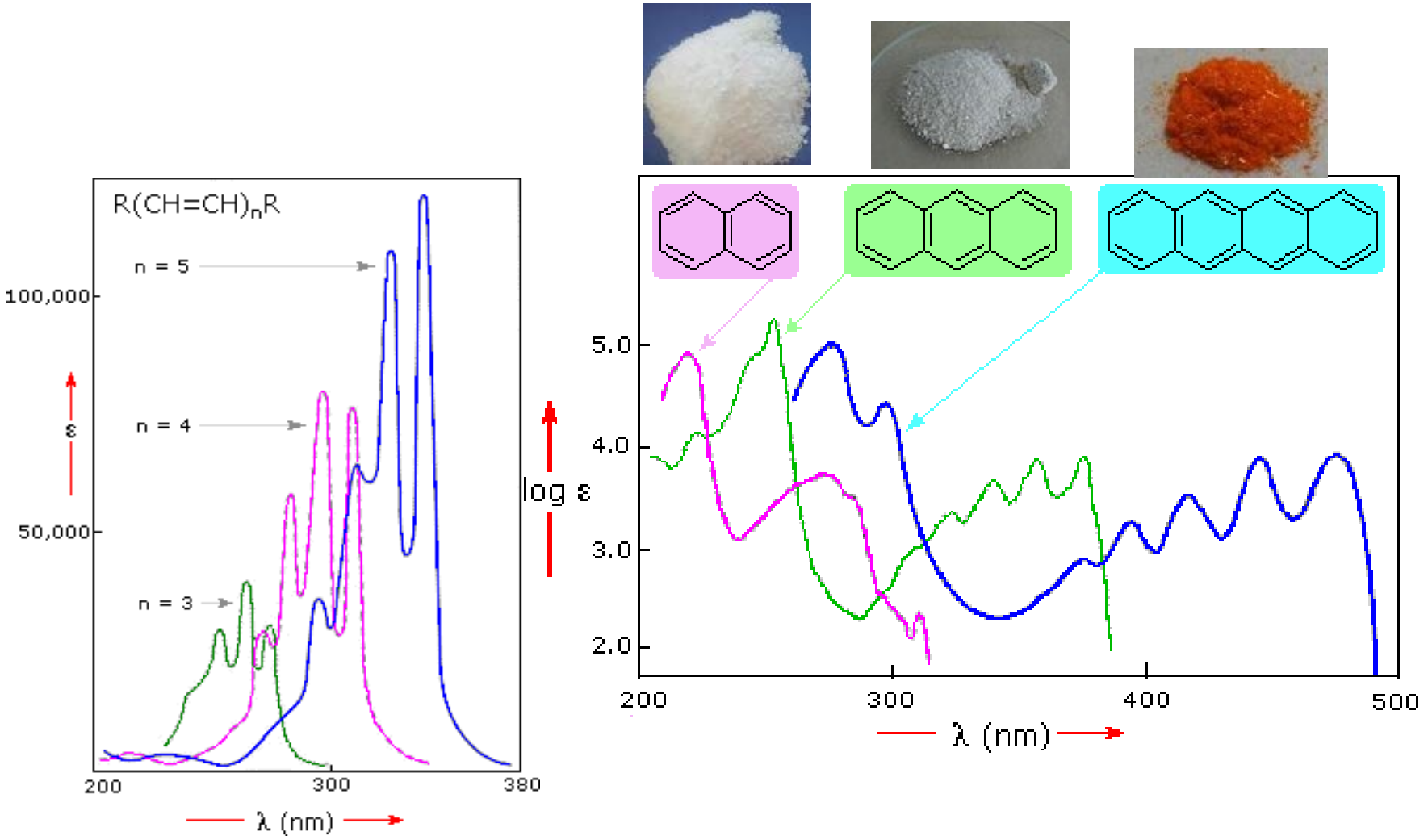


Both molecules are **extended π systems (alternating double and signle bonds)**. The many *p* orbitals on the sp^2 carbon atoms contribute to many π MOs. As a consequence, the molecules have many π molecular energy levels that become very closely spaced.



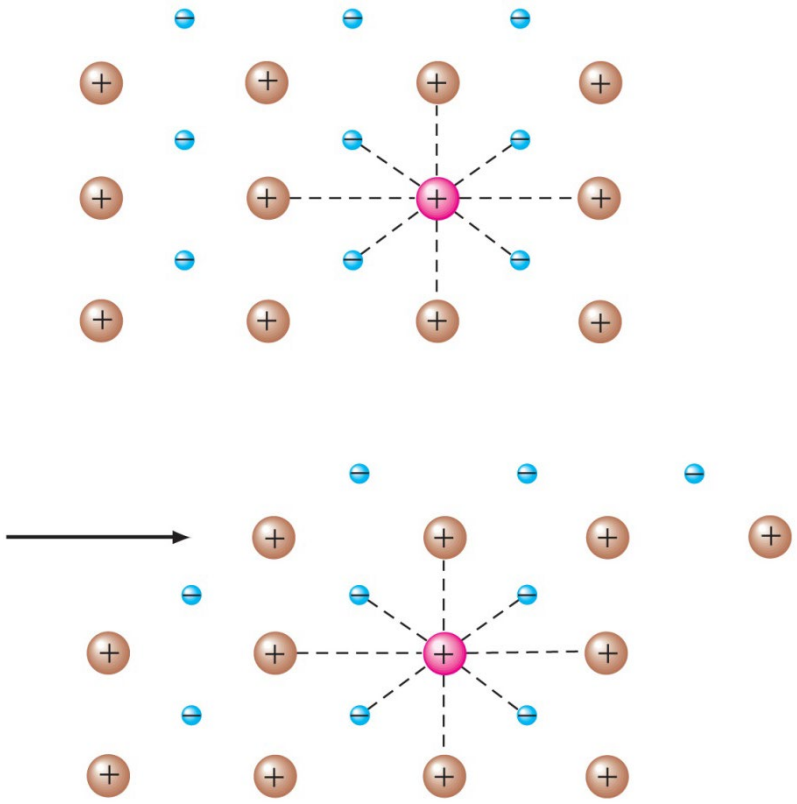
▲ FIGURE 11-35
The formation of π molecular orbitals in a long-chain polyene

The more extensive the π system, the smaller the energy gap between successive levels, hence the smaller the HOMO-LUMO gap. Absorption goes to **longer wavelength**.



As the number of double bonds (**n**) **increases,** the absorption goes to **longer wavelength**.

Bonding in Metals



The electron sea model of metals (e.g. Li)

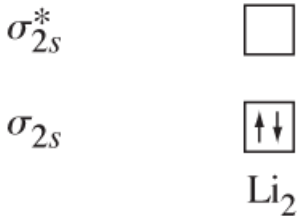
- A network of positive ions is immersed in a "sea of electrons". The electrons are mobile and this accounts for the electrical conductivity of metals.

- At the bottom of the figure, a force is applied to a layer of metal ions (say by hammering), no bonds are broken. The internal structure of the metal remains essentially unchanged, and the sea of electrons rapidly adjusts to the new situation. This explains the ease of deformation of metals (malleability).

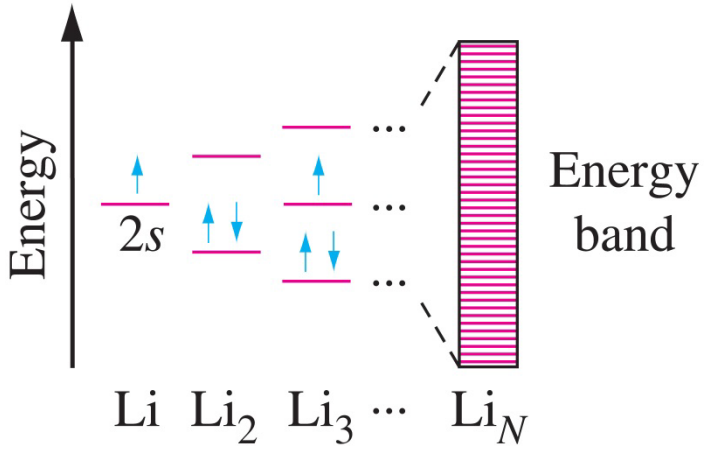
- Electrons in the sea can also absorb photons of visible light and therefore metals are opaque. Electrons at the surface of a metal are able to reradiate at the same frequency light that strikes the surface, which explains the lustrous appearance of metals.

Band Theory

- Recall the formation of molecular orbitals and the bonding between two Li atoms. Each Li atom contributes one 2s orbital to produce two MOs: σ_{2s} and σ^*_{2s}. The set of MOs is half-filled.

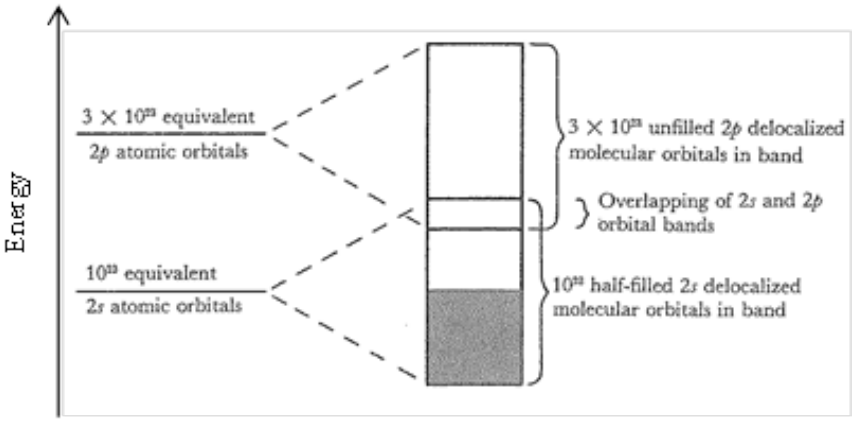


- Extending this to Li_3, three MOs are formed with a total of three electrons. Again, the set of MOs is half-filled. Extend this process further to an enormously large number (N) of atoms – the total number of atoms in a crystal of Li. A set of N molecular orbitals with an extremely small energy separation between each pair of successive levels is resulted. This collection of very closely spaced molecular orbital energy levels is called an ***energy band***.



As more Li atoms are added, the spacing between levels becomes **increasingly smaller**. In an entire crystal of N atoms, the energy levels merge into a band of N closely spaced levels. The lowest $N/2$ levels are filled with electrons (the valence band), and the upper $N/2$ levels are empty (the conduction band).

- In the band, there are *N* electrons occupying, in pairs, *N*/2 MOs of lowest energy. These are the electrons responsible for bonding the Li atoms together. They are valence electrons, and the band in which they are found is called ***a valence band***.



- Because ΔE between the occupied and unoccupied levels in the valence band are so small, electrons can be easily excited from the highest filled levels to the unfilled levels that lie immediately above them in energy. This excitation, which has the effect of producing mobile electrons, can be accomplished by applying a small electric potential difference across the crystal. This is how the band theory explains the ability of metals to conduct electricity.

- The essential feature for electrical conductivity is ***an energy band that is only partly filled with electrons***. Such an energy band is called a *conduction band*. In lithium, the 2*s* band is both a valence band and a conduction band (Fig. 11-38a).

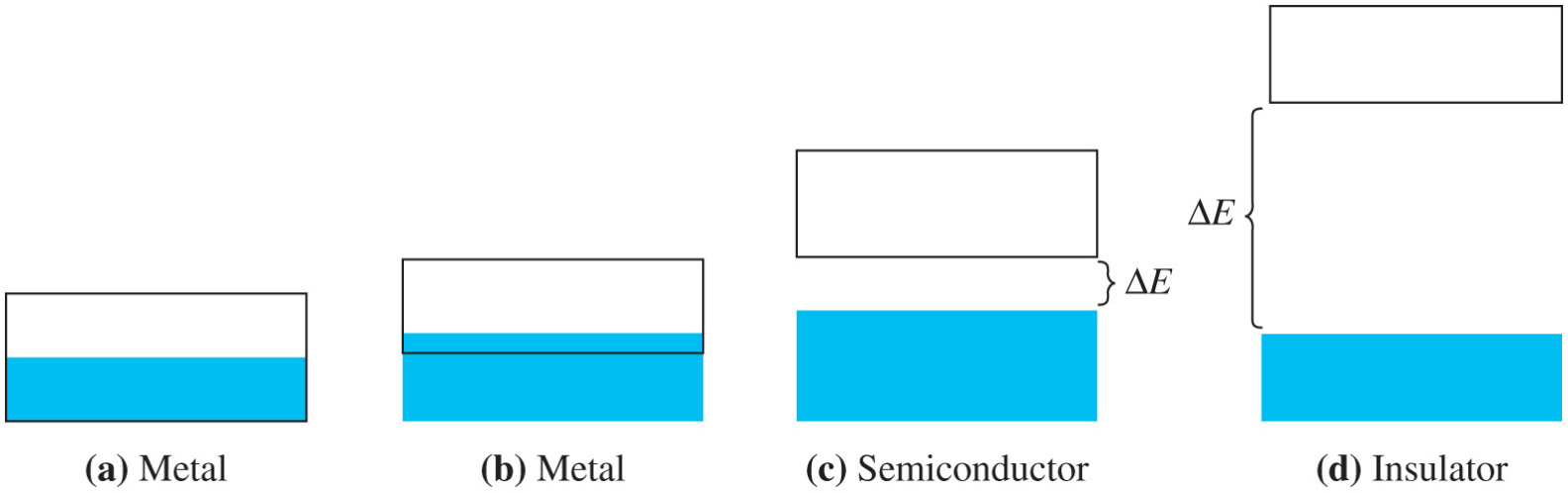


Figure 11-38 Metals, semiconductors, and insulators as viewed by band theory

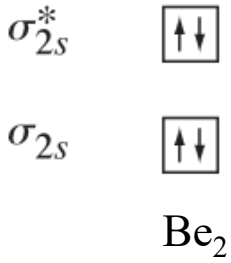
(a) In some metals, the valence band (blue) is only partially filled (e.g. the half-filled 3*s* band in Na). The valence band also serves as a conduction band (outlined in black).

(b) In other metals, the valence band is full, but a conduction band overlaps it (e.g. the empty 2*p* band of Be overlaps the full 2*s* valence band).

(c) In a semiconductor (e.g. silicon, germanium), the valence band is full and the conduction band is empty. The energy gap (ΔE) between the two is small enough, however, that some electrons make the transition between them just by acquiring thermal energy.

(d) In an insulator (e.g. diamond, silica), the valence band is filled with electrons and a large energy gap (ΔE) separates the valence band from the conduction band. Few electrons can make the transition between bands, and the insulator does not conduct electricity.

- Consider *N* atoms of beryllium ($1s^2 2s^2$), it is expected that the band formed from 2*s* atomic orbitals to be filled – *N* molecular orbitals and 2*N* electrons.



- ***How to account for the good electric conductivity of Beryllium by band theory?***

  - While 2*s* orbitals are being combined into a 2*s* band, 2*p* orbitals also combine to form an *empty* 2*p* band. The lowest levels of the 2*p* band are at a *lower* energy than the highest levels of the 2*s* band. The bands overlap (Fig.11-38b). As a consequence, empty MOs are available to the valence electrons in beryllium.

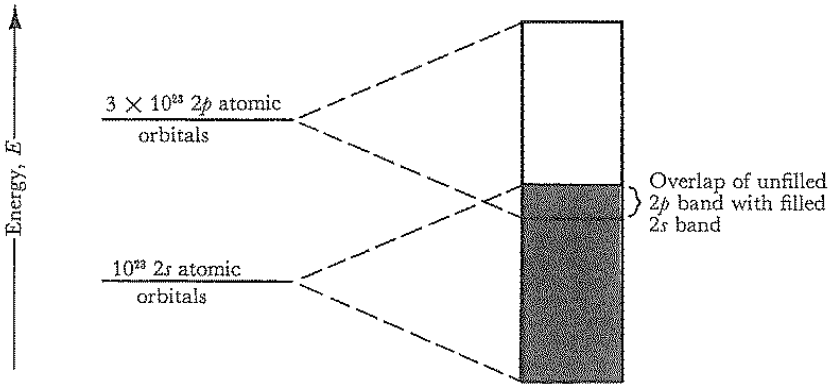
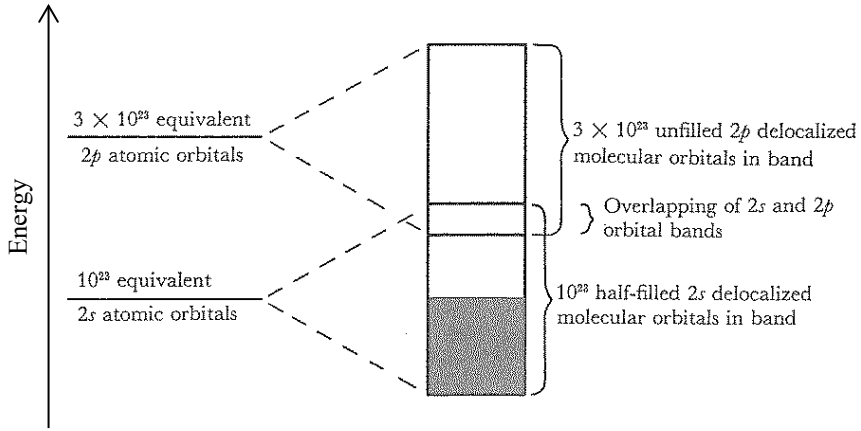
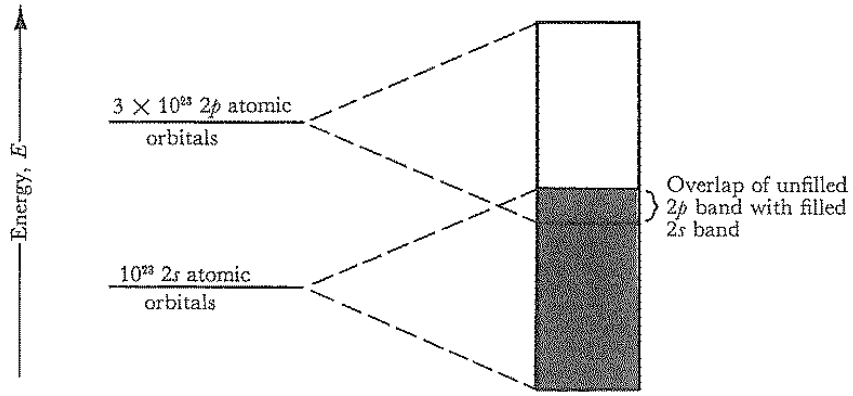


Figure a)
Delocalised MO bands in **lithium**



- The original 2*s* and 2*p* AOs are so close in energy that the MO bands overlap.
- Lithium has one electron in every 2*s* AO, hence only half as many electrons as can be accommodated in the delocalized MO band.
- There are unfilled energy states an infinitesimal distance above the highest-energy filled state, so an infinitesimal energy is required to excite an electron and send it moving through the metal. Thus lithium is a conductor.

Figure b)
Delocalised MO bands in **beryllium**



- A beryllium atom has enough electrons (two) to fill its 2*s* orbital, so beryllium metal has enough electrons to fill its 2*s* delocalized MO band.
- If the 2*s* and 2*p* bands did not overlap, beryllium would be an insulator, because an appreciable amount of energy would be required to make electrons flow in the solid.
- But with the band overlap shown here, an infinitesimal amount of energy excites the electrons to the 2*p* band orbitals, and electrons flow.

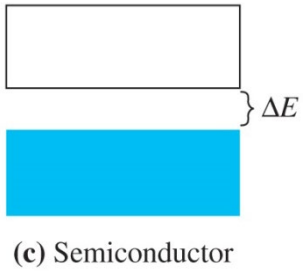
Physical Properties of Metals in terms of the Band Theory

1. Metals have good electrical conductivity.

2. Metals have a lustrous appearance because the mobile electrons can absorb a wide range of wavelengths of radiant energy as they jump to higher energy levels. Then they emit photons of visible light and fall back to lower levels. This ability to re-radiate photos accounts for the shininess.

3. Metals are malleable or ductile (or both). A crystal of a metal is easily deformed when a mechanical stress is applied to it. All of the metal ions are identical, and they are imbedded in a “sea of electrons”. As bonds are broken, new ones are readily formed with adjacent metal ions because successive energy levels are so close.

Semiconductors



(c) Semiconductor

- Semiconductor materials are widely used in modern electronics (e.g. light-emitting diodes, transistors, and solar cells).
- What determines the electronic properties of a semiconductor is the energy gap (band gap) between the valence band and the conduction band (Fig. 11-38c). In some materials, such as **cadmium yellow (CdS),** this gap is of a fixed size. These materials are called *intrinsic semiconductors*. When white light interacts with the semiconductor, electrons are excited to the conduction band. CdS absorbs violet light and some blue light, but other frequencies contain less energy than is needed to excite an electron above the energy gap. The frequencies that are not absorbed are reflected, and the color we see is **yellow**.

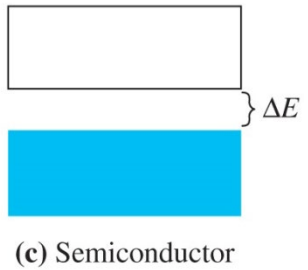




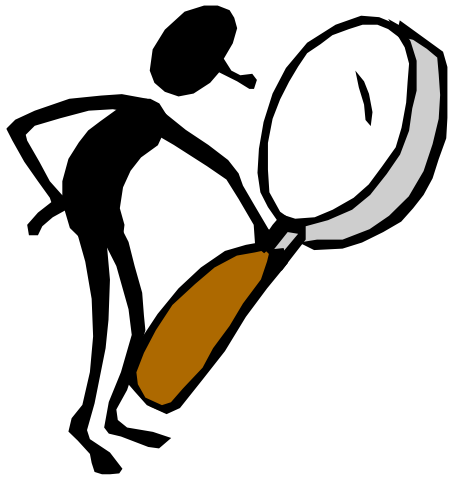
- Some semiconductors, such as **gallium arsenide (GaAs) and lead(II) sulfide (PbS),** have a sufficiently small band gap that all frequencies of visible light are absorbed. There is no reflected visible light, and the materials have a black color.

Semiconductors

- In a semiconductor, such as silicon or germanium, the filled valence band and empty conduction band are separated by only a small energy gap (Fig. 11-38c). Electrons in the valence band may acquire enough thermal energy to jump to a level in the conduction band. The greater the thermal energy, the more electrons can make the transition. In this way, band theory explains the observation that the electrical conductivity of semiconductors increases with temperature.

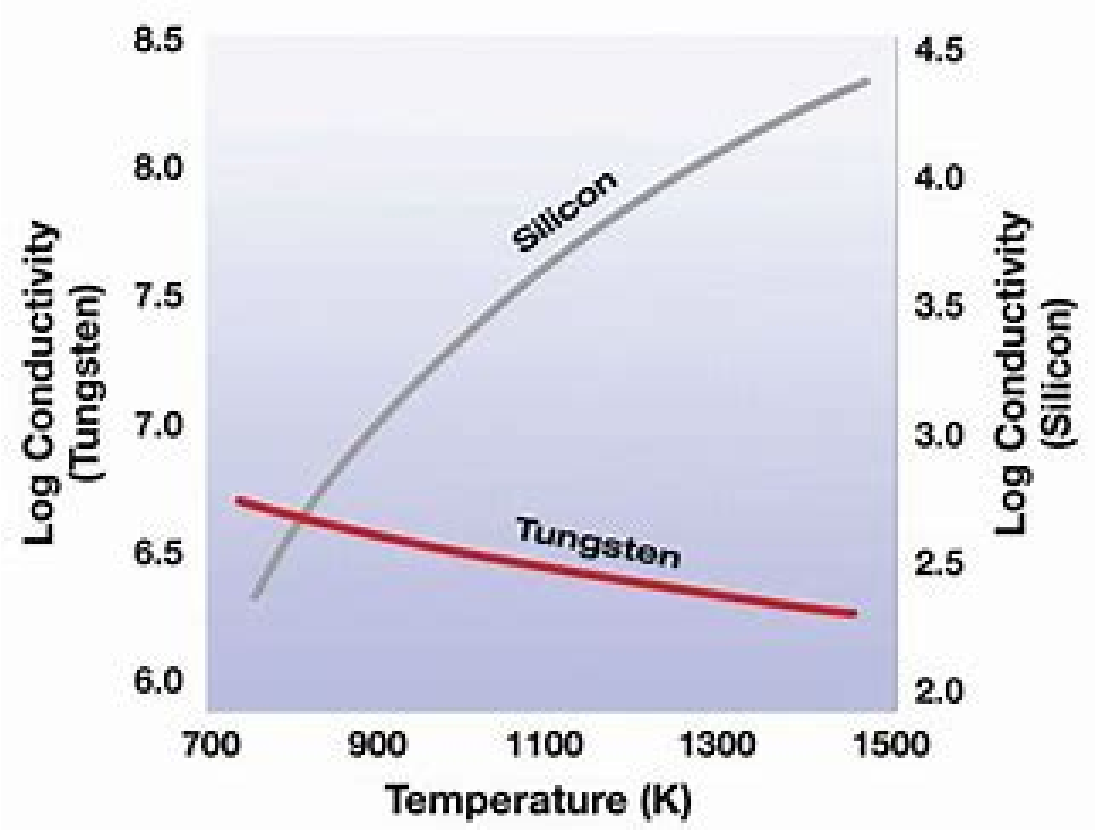


(c) Semiconductor



You shall learn more on semiconducting materials (e.g. p- and n-type semiconductors) if you will take higher level CHEM courses in future.

The electrical conductivity of semiconductors increases with temperature



Quick Summary on Metallic Conductors and Semiconductors

- A metallic conductor is a substance with an electric conductivity that ***decreases*** with increasing temperature. (sea of electron)

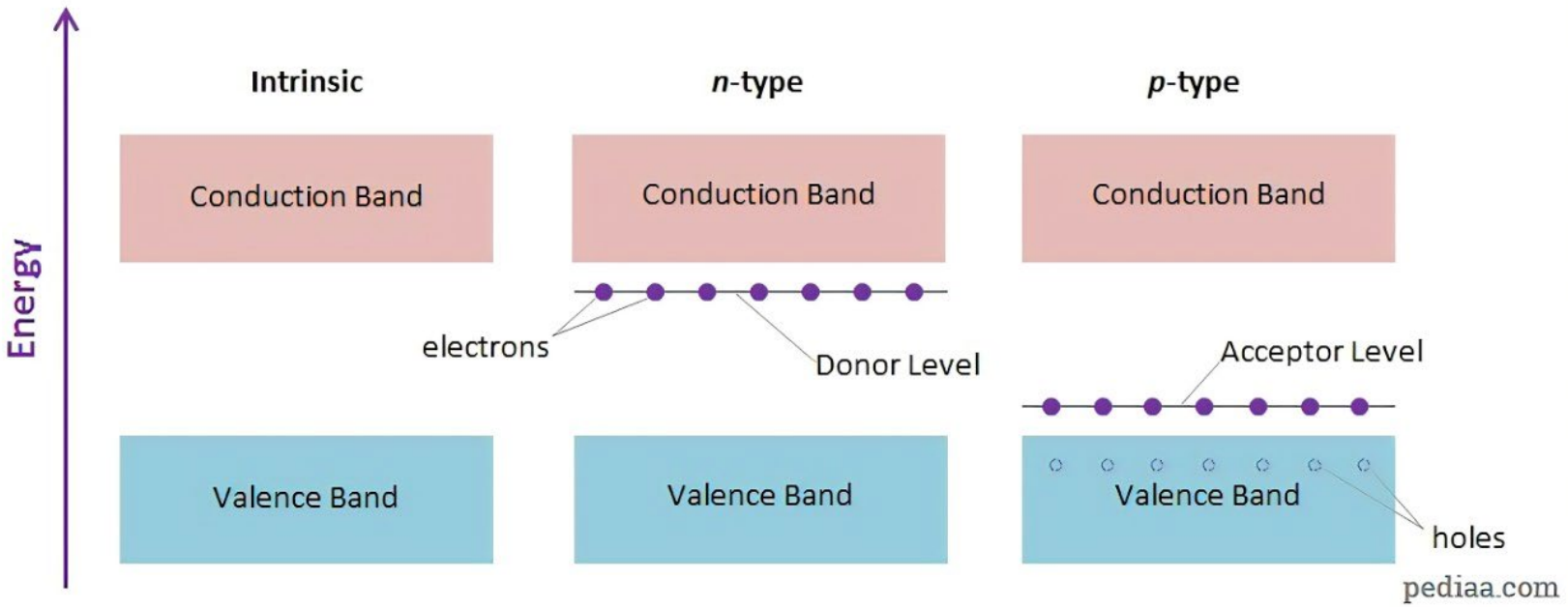
- A semiconductor is a substance with an electric conductivity that ***increases*** with increasing temperature. (band theory)

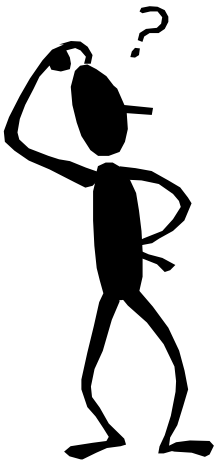
- For semiconductors, there are two types.
  - In an intrinsic semiconductor, the band gap is so small that some electrons make the transition between them just by acquiring thermal energy.

  - In an extrinsic semiconductor, some impurities have been added to them to increase their conductivity. This is called doping.

Quick Summary on Metallic Conductors and Semiconductors

- In an extrinsic semiconductor, some impurities have been added to them to increase their conductivity. This is called doping.





How can Molecular Orbital Theory be applied to Bonding in Metals?

Molecular orbital theory, in the form called **band theory**, can be applied to metals, semiconductors, and insulators (Fig. 11-38). Band theory explains thermal and electric conductivity, the ease of deformation, and the characteristic luster of metals. It also explains the colors of semiconductors and the fact that their electric conductivities increase with temperature.



Class Practice 7

Magnesium is an excellent electrical conductor even though it has a full 3*s* subshell with the electron configuration: [Ne]3s^2. Use band theory to explain why magnesium conducts electricity.

Solution:

Although the band that results from the combination of 3*s* orbitals in magnesium metal is full, there is another empty band that overlaps this 3*s* band. This empty band results from the combination of 3*p* orbitals. Although it does not overlap the 3*s* band in the Mg_2 molecule or in atom clusters, once the metal crystal has grown to visible size, the overlap is complete.